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FARMEKOX: SMART CROP ADVISORY SYSTEM FOR SMALL AND MARGINAL FARMERS

A PROJECT REPORT

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IN

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "FARMEKOX: SMART CROP ADVISORY SYSTEM FOR SMALL AND MARGINAL FARMERS" is a bonafide work of "NAVEEN M(20221CSG0153), AVINASH N(20221LCG0004), KURUBA ARUN KUMAR (20221LCG0005)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND TECHNOLOGY during 2025-2026

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ABSTRACT

Agriculture is a major source of livelihood for a large population in India, especially for small and marginal farmers. However, farmers often face challenges such as lack of timely information, improper crop selection, pest and disease attacks, and limited access to market data. These issues result in reduced productivity and financial instability.

The proposed system, **FarmEkoX: Smart Crop Advisory System**, aims to address these challenges by providing an integrated digital platform that supports farmers with real-time, data-driven decision-making. The system is designed as a Progressive Web Application (PWA), ensuring accessibility even in low-bandwidth rural environments.

The platform incorporates advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) to deliver intelligent advisory services. A key feature of the system is crop recommendation based on location, season, and environmental conditions. Additionally, the system provides weather-based irrigation alerts, fertilizer guidance through barcode scanning, and real-time market price updates.

The AI advisory module, integrated using modern generative AI technologies, allows farmers to interact through text and voice in regional languages such as Kannada and English. This improves usability and bridges the digital literacy gap among rural users. The system also includes features like expert consultation and nearby resource identification, enhancing overall farmer support.

The FarmEkoX system is developed using modern web technologies including React.js, Node.js, and MongoDB, ensuring scalability and performance. By combining multiple agricultural services into a single platform, the system improves decision-making, reduces resource wastage, and increases productivity.

Overall, the proposed solution contributes to sustainable agriculture by promoting efficient resource usage, improving farmer income, and supporting digital transformation in rural farming communities.

Table of Content

Sl. No.	Title	Page No.
	Declaration	iii.
	Acknowledgement	iv.
	Abstract	v.
	List of Figures	viii.
	List of Tables	ix.
	Abbreviations	x.
1.	Introduction 1.1 Background 1.2 Statistics of project 1.3 Prior existing technologies 1.4 Proposed approach 1.5 Objectives 1.6 SDGs 1.7 Overview of project report	1-9
2.	Literature review	10-12
3.	Methodology	13-22
4.	Project management 4.1 Project timeline 4.2 Risk analysis 4.3 Project budget	23-26
5.	Analysis and Design 5.1 Requirements 5.2 Block Diagram 5.3 System Flow Chart 5.4 Hardware Components and Specifications 5.5 Software Components and Requirements 5.6 Functional Modules 5.7 Datasets and Data Resources	27-37

6.	Hardware,Softwareand Simulation 6.1 Hardware 6.2 Software development tools 6.3 Software code	38-41
7.	Evaluation and Results 7.1 Test points 7.2 Test plan 7.3 Test result 7.4 Insights	42-44
8.	Social, Legal, Ethical, Sustainability and Safety Aspects 8.1 Social aspects 8.2 Legal aspects 8.3 Ethical aspects 8.4 Sustainability aspects 8.5 Safety aspects	45-47
9.	Conclusion	48-49
	References	50
	Base Paper	50
	Appendix	51-55

List of Figures

S. No.	Figure Caption	Page No.
Fig 1.6	Sustainable Development Goals (SDG)	08
Fig 3.1	The V-Model Methodology	13
Fig 3.3	Block diagram	20
Fig 4.1	Gantt Chart	24
Fig 5.1	Functional Block Diagram	30
Fig 5.3	System Flow Chart	32
Fig 5.3	Web Application Interface	34
Fig 6.1	Agro-Mentor Homepage	38
Fig 6.2	AI Disease Detection Module Interface	38
Fig 6.3	Expert Consultation Model Interface	38
Fig 6.4	Farmer Community Module Interface	39
Fig 9.1	Home page	52
Fig 9.2	Ai Chat	52
Fig 9.3	Direct Sales	53
Fig 9.4	Product Identification	53
Fig 9.5	Crop Recommendation	53
Fig 9.6	Expert Connect	54
Fig 9.7	Market Watch	55
Fig 9.8	Dashboard	55
Fig 9.9	Nearby resources	55
Fig 9.10	Similarity Report	56
Fig 9.11	AI Detection Report	57

List of Tables

Table ID	Table Caption	Page No.
Table 1.2	Statistics of Project	03
Table 2.5	Reviewed literature table	12
Table 4.1	Timeline	23
Table 4.2	Implementation phase timeline	24
Table 4.3	Pestle Analysis	25
Table 4.4	Project Budget	26
Table 5.1	Summarizing Requirements	29
Table 5.5	Components and Requirements	35
Table 5.6	Functional Models	36
Table 7.3	Test Results	43

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CSS	Cascading Style Sheet
DBMS	Database Management System
IDE	Integrated Development Environment
ML	Machine Learning
NLP	Natural Language Processing
API	Application Programming Interface
PWA	Progressive Web Application
SDLC	Software Development Life Cycle
UI	User Interface
UX	User Experience
GPS	Global Positioning System
IoT	Internet of Things
PGSS	Presidency School of Computer Science

Chapter 1

Introduction

The agricultural sector in India is currently undergoing a paradigm shift, as traditional methods of cultivation encounter the twin pressures of unprecedented climate change and a rapidly evolving digital landscape. Agriculture and allied activities remain the backbone of the Indian economy, contributing approximately 16% to the national Gross Domestic Product (GDP) and supporting the livelihoods of nearly 46.1% of the population as of the 2024-25 fiscal year. Despite record foodgrain production reaching 357.73 million metric tonnes (MMT) in the latest reporting cycle, the average smallholder farmer continues to struggle with issues of low productivity, high input costs, and environmental volatility. The majority of Indian farm households—approximately 86%—are classified as small or marginal, managing landholdings of less than two hectares, with an average size as low as 0.5 hectares per household.

The primary obstacle to modernizing this sector is a pervasive "Decision Gap." Farmers often rely on traditional guesswork, informal advice, or seasonal intuition for critical management decisions such as crop selection, pest control, and nutrient management. This absence of personalized, data-driven insights frequently leads to the late detection of crop diseases, suboptimal irrigation practices, and the overuse of fertilizers—factors that not only diminish yields but also cause significant environmental damage, including soil degradation and groundwater contamination. In regions like Punjab, which serves as India's breadbasket, the situation is particularly dire; the water table is declining at an alarming rate of 1.7 feet per year, and recent floods in 2025 destroyed ready-to-harvest crops across nearly 700,000 hectares, leading to economic losses exceeding 7,500 crore for rice alone.

FarmEkoX: Smart Crop Advisory System is designed to bridge this information asymmetry by providing a unified, AI-powered digital ecosystem tailored to the needs of small and marginal farmers. By integrating cutting-edge technologies including React.js, Next.js, Node.js, and the Google Gemini API, the platform delivers real-time, site-specific, and multilingual agricultural guidance. The system transforms a standard smartphone into a precision agriculture tool, offering features such as AI voice assistance in local languages, barcode-based fertilizer identification, and predictive rain alerts.

1.1 Background and motivation

Indian agriculture has historically functioned through traditional extensions, where field visits by agricultural officers were the primary source of expert knowledge. However, these conventional services are often hampered by geographic constraints, resource limitations, and a low officer-to-farmer ratio. Consequently, many farmers remain ill-equipped with up-to-the-minute data on crop platform serves as a lever to raise productivity, keep profitability rates high, and help seed long-term resilience in the Indian agricultural industry through an affordable and accessible digital solution.

health or local weather patterns. Research indicates that global crop production losses due to pests and diseases range between 20% and 40%, highlighting the critical need for immediate, on-site diagnostic tools.

The digital divide in rural India further complicates this issue. While the country boasts over 958 million active internet users, with 57% of them residing in rural areas, access does not necessarily equate to productive utility. A "second-level digital divide" exists, characterized by a lack of digital literacy and the complexity of existing interfaces that alienate older or semi-literate farmers. Furthermore, rural women, who perform a significant portion of agricultural labor, are 41% less likely to use mobile internet than their male counterparts due to social norms and skill gaps. FarmEkoX aims to address these systemic inequities by utilizing a voice-first, multilingual Progressive Web App (PWA) design that functions effectively even on shared devices and low-bandwidth connections.

1.2 Statistics of Project

Understanding the regional and national context through statistical analysis is vital for identifying the scope of the FarmEkoX project. The following table summarizes key agricultural and digital metrics relevant to the platform's deployment.

dollars to farmers annually and low-income instability is a result of those who depend on farming.

Then, we have a digital divide: a majority of people can afford and own a mobile phone, but very few will think of using an agricultural app. Some of the issues are language, and complex UI, and the compulsory downloading of several apps for various services.

Thirdly, the project is making a difference: the Agro-Mentor AI Platform is coming up with more than just good results:

- More farmers can be helped
- Over 200 diseases have been identified
- Success rate stands at 94%
- All reginal lancuages can be updated

These numbers are an eye-opener to the enormous potential of a single tech-driven agriculture platform powered by AI and big data in transforming farmers' practices and increasing their income levels.

Metric Description	Statistical Value	Source
Rural Active Internet Users	548 Million (57% of total)	Internet and Mobile Association of India (IAMAI) Digital Report 2025
Marginal Farmer Households	86% of total Indian farms	Ministry of Agriculture and Farmers Welfare, Government of India
Average Monthly Farm Household Income	Rs 13,661 (< $160)	National Sample Survey Office (NSSO) Situation Assessment Survey
Crop Loss due to Pests and Diseases	20% to 30% of potential yield	Food and Agriculture Organization (FAO), United Nations
Digital Payment Adoption among Farmers	43% (skyrocket from 11% in 2022)	Reserve Bank of India (RBI) Digital Payment Report
Rural Female Internet Usage Rate	24.6% (vs 48.7% for men)	Internet and Mobile Association of India (IAMAI)
Paddy Procurement Decline in Punjab (2025)	15% reduction from previous year	Food Corporation of India (FCI) Procurement Data
Yield Gap in Coconut (Karnataka)	36.16% (potential vs actual)	Indian Council of Agricultural Research (ICAR)

Table 1.2 Statistics of Project

These figures underscore the massive opportunity for a single, tech-driven platform to transform farming practices. The jump in digital payments suggests that farmers are becoming comfortable with digital transactions (UPI), yet they still lack specialized tools for agronomic decision-making.

1.3 Prior Existing Technologies

Several digital solutions currently exist in the Indian agritech landscape, yet none provide a truly holistic ecosystem for smallholder needs. Applications such as Plantix, AgriApp, and IFFCO Kisan have made significant strides but face persistent limitations:

- **Fragmentation:** Most existing platforms are feature-oriented, forcing farmers to download multiple apps for weather, market prices, and disease identification.
- **Language and Literacy Barriers:** Although some apps provide regional support, they often rely on complex text menus. For example, 40.83% of m-Kisan users reported problems with foreign languages and technical jargon.
- **Connectivity and Infrastructure:** High-resolution image analysis for disease detection often fails in fields where 4G/5G signals are spotty. PWAs like Agro-Mentor and FarmEkoX are emerging to solve this via local caching.
- **Lack of Localization:** General apps often fail to provide specific mandi prices or dialects unique to regions like the Punjab or rural Karnataka.

1.4 Proposed Approach

1.4.1 Mobile-First Progressive Web Application (PWA) Design

The FarmEkoX system adopts a mobile-first strategy by developing a Progressive Web Application (PWA) that ensures seamless accessibility across a wide range of devices, including low-end smartphones commonly used by small and marginal farmers. This approach eliminates the need for downloading heavy native applications and allows the system to function efficiently even under low-bandwidth conditions. The PWA architecture supports offline capabilities through caching mechanisms, enabling farmers to access essential features such as saved recommendations and previously fetched data without continuous internet connectivity. This design choice directly addresses the infrastructural limitations prevalent in rural areas and ensures inclusivity for users with limited digital resources.

1.4.2 Generative AI-Based Advisory System

At the core of the proposed system lies a powerful Generative AI engine that provides intelligent, real-time agricultural guidance. By leveraging advanced AI models, the system can understand and respond to farmer queries in natural language through both text and voice interfaces. The advisory system is designed to be bilingual or multilingual, supporting regional languages such as Kannada and Punjabi alongside English. This ensures that farmers can

interact with the system comfortably in their native language. The AI module processes queries related to crop diseases, pest management, fertilizer usage, and weather conditions, delivering context-aware and personalized recommendations. This significantly reduces reliance on traditional guesswork and enhances decision-making accuracy.

1.4.3 Precision Farming and Resource Management Tools

The proposed approach integrates precision agriculture techniques to optimize resource utilization. One of the key features is a barcode-based fertilizer identification system, which allows farmers to scan product packaging and instantly retrieve detailed nutrient composition (such as N-P-K values). Based on this information, the system provides accurate dosage recommendations tailored to specific crops and land areas. Additionally, geolocation-based weather monitoring enables real-time rain alerts, allowing farmers to avoid unnecessary irrigation and conserve water. These tools collectively contribute to reducing input costs, minimizing environmental impact, and improving overall farm productivity.

1.4.4 Smart Crop Recommendation and Planning Engine

FarmEkoX incorporates a smart crop recommendation module that assists farmers in selecting the most suitable crops based on various parameters such as geographic location, soil conditions, seasonal trends, and historical weather data. By analyzing these factors, the system generates data-driven suggestions that maximize yield potential and minimize risk. This planning engine also supports crop rotation strategies and long-term agricultural planning, helping farmers make informed decisions before the start of each cultivation cycle. As a result, farmers can transition from traditional intuition-based practices to a more scientific and predictive approach to farming.

1.4.5 Market Intelligence and Resource Accessibility

To enhance the economic well-being of farmers, the proposed system integrates real-time market intelligence features. Farmers can access up-to-date mandi (market) prices for various crops, enabling them to choose the best time and location to sell their produce. Additionally, the platform includes a resource locator that helps users identify nearby agricultural input stores, service providers, and marketplaces using location-based services. This feature reduces dependency on intermediaries and empowers farmers with better bargaining power. By combining market insights with advisory services, FarmEkoX ensures a holistic approach to agricultural decision-making, covering both production and post-harvest stages.

1.5 Objectives

The objectives of the proposed system Farmekox – Smart Crop Advisory System are defined to address key aspects such as behaviour, analysis, system management, security, and deployment. These objectives are measurable and demonstrable through the implementation of the system.

i. Behaviour Objective:

To develop a responsive crop advisory system that provides suitable crop recommendations based on user inputs such as location, season, and climatic conditions. The system should dynamically adjust its output according to real-time inputs and guide farmers in selecting appropriate crops and farming practices.

ii. **Analysis Objective:**

To design and implement a data-driven analysis mechanism that processes multiple inputs including weather data, crop information, and farmer queries. The system should utilize AI models and rule-based logic to generate accurate and personalized recommendations for better decision-making.

iii. **System Management Objective**:

To build an integrated platform that efficiently manages multiple agricultural services such as crop recommendation, crop calendar, weather alerts, fertilizer information, and market price updates within a single application. The system should ensure smooth navigation and centralized access through a user-friendly dashboard.

iv**. Security Objective**:

To ensure secure handling of user data by implementing authentication and safe data transmission mechanisms. The system should protect user inputs such as location and queries using secure backend services and maintain basic data privacy standards.

v. **Deployment Objective**:

To deploy the application as a scalable web-based platform accessible on both mobile and desktop devices. The system should be lightweight, responsive, and capable of running efficiently in rural environments with limited resources and connectivity.

1.6 Sustainable Development Goals (SDGs)

The Farmekox – Smart Crop Advisory System aligns with several United Nations Sustainable Development Goals (SDGs) by addressing real-world agricultural challenges using technology.

Agriculture plays a major role in India's economy, and improving farming practices directly contributes to sustainability, food security, and rural development. The objectives and features of this project support the following SDGs:

SDG 2: Zero Hunger

The system helps farmers improve crop productivity by providing accurate crop recommendations, fertilizer guidance, and timely advisory. By reducing guesswork and promoting scientific farming methods, the project contributes to increased food production and better yield quality. In this project, we observed that farmers can make better crop choices when guided by seasonal and climatic data, which directly supports food security.

SDG 1: No Poverty

Many small and marginal farmers face financial instability due to poor crop planning and low market awareness. The inclusion of features like Market Watch (mandi prices) and Direct Sales enables farmers to make better selling decisions and potentially increase their income. By reducing unnecessary expenses such as overuse of fertilizers and water, the system also helps in lowering input costs.

SDG 12: Responsible consumption and Production

The fertilizer recommendation and irrigation alert modules promote efficient use of resources. For example, the system provides rain alerts suggesting farmers skip irrigation when rainfall is expected. This reduces water wastage and encourages sustainable farming practices. Similarly, the fertilizer calculator helps avoid excessive use of chemicals, which can harm soil health.

SDG 9: Industry, Innovation and Infrastructure

The project integrates modern technologies such as AI (Google Gemini API), cloud services (Firebase), and web technologies (React.js). This promotes digital innovation in agriculture and contributes to building smart agricultural infrastructure. The use of AI advisory in local languages also shows how innovation can be adapted for rural users.

SDG 13: Climate Action

Weather-based recommendations and alerts help farmers adapt to changing climatic conditions. By providing real-time weather forecasts and irrigation advice, the system supports climate-resilient farming practices. This is especially important in regions where unpredictable weather affects crop yield.

SDG 10: Reduced Inequalities

One of the important aspects of this project is the support for local languages such as Kannada.

Many existing platforms are not accessible to farmers due to language barriers. By providing voice and chat-based advisory in regional languages, the system ensures inclusivity and reduces the digital divide between urban and rural users.

Fig 1.6 Sustainable development goals [1]

1.7 Overview of project report

This project report is organized into multiple chapters, each focusing on a specific aspect of the development and implementation of the *Farmekox – Smart Crop Advisory System*. The structure is designed to provide a clear understanding of the problem, the proposed solution, and the overall system development process.

- **Chapter 1: Introduction** presents the background of the project, the need for a smart agricultural advisory system, and the limitations of existing solutions. It also includes the objectives, alignment with Sustainable Development Goals (SDGs), and a general overview of the project.
- **Chapter 2: Literature Review** discusses various research works and existing agricultural technologies related to crop recommendation, AI-based advisory systems, and smart farming solutions. It highlights the gaps in current systems and provides justification for the proposed approach.
- **Chapter 3: Methodology** explains the development approach used in this project. It describes the selected software development model, phases involved, and how the system was designed and implemented step by step. The workflow of the system is also discussed in detail.
- **Chapter 4: Project Management** focuses on the planning and execution aspects of the project. It includes the project timeline (Gantt chart), risk analysis, and budget

estimation. This chapter explains how the project was managed effectively within the given time and resources.

- **Chapter 5: Analysis and Design** provides a detailed description of system requirements, architecture, block diagrams, and flowcharts. It explains how the system is structured and how different modules interact with each other to deliver the desired functionality.
- **Chapter 6: Implementation** describes the actual development of the system, including frontend, backend, database design, and AI integration. It explains how different technologies such as React.js, Firebase, and Google Gemini API were used to build the application.
- **Chapter 7: Results and Evaluation** presents the outcomes of the project along with test cases, observations, and performance evaluation. It analyzes how effectively the system meets the defined objectives.
- **Chapter 8: Social, Ethical, Legal, and Sustainability Aspects** discusses the broader impact of the project on society, including ethical considerations, legal aspects, and sustainability benefits.
- **Chapter 9: Conclusion** summarizes the overall work carried out in the project and highlights key findings. It also provides insights into possible future enhancements and improvements. Finally, the report includes **References** in IEEE format and an **Appendix** section containing supplementary information such as additional data, screenshots, and supporting documents.

  This structured approach ensures that all aspects of the project are clearly documented and easy to understand for evaluation and future reference.

Chapter 2

Literature Review

The theoretical foundation of FarmEkoX is built upon a synthesis of academic research in Computer Vision, Natural Language Processing (NLP), and rural sociology. The literature review highlights the transition from traditional extension methods to AI-driven "Smart Farming" systems, identifying critical gaps in the delivery of information to smallholder farmers in the Global South.

2.1 Deep Learning for Plant Disease Recognition

The use of Convolutional Neural Networks (CNNs) has revolutionized the field of plant pathology. Kamilaris and Prenafeta-Boldú (2018) conducted a landmark survey that demonstrated the superiority of deep learning over traditional machine learning for image-based agricultural tasks, noting its exceptional feature extraction capabilities for complex leaf patterns. Sladojevic et al. (2016) further validated this by showing that CNN models could classify a wide range of diseases with high accuracy even in the presence of noise and poor image resolution, provided they were trained on large, annotated datasets like "Plant Village".

However, "in-the-wild" classification remains a significant challenge. Ferentinos (2018) pointed out that models trained in controlled laboratory environments often see a 10-15% drop in accuracy when deployed in field conditions with varying lighting, complex backgrounds, and multiple symptoms on a single leaf. This has led to the development of ensemble CNN architectures and data augmentation techniques—such as rotation, scaling, and color jittering—to improve robustness. FarmEkoX adopts these principles by fine-tuning models on locally captured images from the Punjab and Karnataka regions to ensure regional relevance.

2.2 Generative AI and Multilingual Voice Interfaces

The emergence of Large Language Models (LLMs) like Google Gemini has opened new avenues for conversational agriculture. Previous generations of chatbots relied on rigid, rule-based decision trees that often failed to understand the nuance of farmer queries. Modern GenAI systems can process vast amounts of unstructured scientific data and provide context-aware responses in natural language.

A critical finding in recent studies is the importance of local vernacular. Research by Suresh Babu and Praveen Ramadugo (2026) on AI voice agents for Telugu farmers emphasized that "textbook-like" responses are often perceived as untrustworthy or confusing. Their research utilized a dataset of over 5,000 village-level vernacular terms to train models that "speak" the language of the farmer, a technique essential for building trust in digital advisory. FarmEkoX implements these findings by using the Genkit framework to ground Gemini's responses in localized agricultural databases.

2.3 Socio-Economic Barriers and the Digital Divide

Technological capability does not automatically translate to adoption. The "Agricultural Digital Divide" research highlights that even when internet penetration reaches 60%, a "usage gap" persists where digital tools are used for entertainment rather than productivity. Factors such as gender, age, and education levels significantly impact digital participation. For instance, rural women in India are 41% less likely to use mobile internet than men, often due to social constraints and a lack of access to personal devices.

Studies by Kalusopa (2005) and more recent works in 2024 indicate that complex user interfaces and the high cost of data remain the primary deterrents for marginal farmers. Furthermore, 60% of farmers still lack access to advanced agricultural technology information. This justifies the FarmEkoX decision to use a PWA (Progressive Web App) architecture, which offers a native-app-like experience without the heavy download size and requirement for high-end hardware.

2.4 Precision Resource and Nutrient Management

The economic impact of resource wastage is a significant driver for precision agriculture. In India, 20-30% of potential crop production is lost to pests and diseases, and the overuse of pesticides has led to environmental health disasters like the endosulfan crisis in Kerala. Precision tools can reduce input costs by 20% while increasing yield by 30%.

Barcoding technology is emerging as a solution for traceability and nutrient management. The global "Sunrise 2027" initiative by GS1 aims to transition retail from 1D to 2D barcodes, allowing products to store rich metadata such as nutrient composition and expiry dates. This is critical for fertilizer management, where incorrect dosages can lead to both economic loss and

soil toxicity. FarmEkoX's inclusion of a barcode scanner for identifying fertilizers directly addresses this emerging global standard.

2.5 Summary of Reviewed Literature

Key Focus Area	Representative Studies	Major Findings and Relevance to FarmEkoX
Deep Learning & CNNs	Kamilaris (2018), Sladojevic (2016)	Validated CNNs for feature extraction and disease recognition from images.
Environmental Robustness	Ferentinos (2018), Ramcharan (2019)	Identified the need for field-level data and augmentation for mobile AI.
Generative AI Chatbots	Digital Green (2023), Babu et al. (2026)	LLM-based voice advisory improves trust and accessibility for low-literacy users.
Digital Divide	IAMAI (2025), Paul et al. (2024)	Emphasized that infrastructure alone is insufficient; UI must be localized and simplified.
Precision Agriculture	Gupta & Verma (2024), Deloitte (2024)	Quantified the yield improvements (+30\%) possible through smart irrigation and nutrient monitoring.

Table 2.5:Reviewed Literature table

The literature consistently points toward the need for a unified platform that combines these disparate technologies into a single, farmer-friendly interface. FarmEkoX fills this gap by integrating state-of-the-art AI with practical precision tools and real-time market data.

Chapter 3

Methodology

3.1 Introduction to V-Model

The **V-Model**, also known as the *Verification and Validation Model*, is a structured software development approach where each phase of development is directly associated with a corresponding testing phase. This model is represented in the shape of the letter "V", where the left side consists of requirement analysis, system design, and implementation, while the right side includes testing activities such as unit testing, integration testing, system testing, and validation.

In the case of implementing the *Farmekox – Smart Crop Advisory System*, the V-Model ensures that every stage of development, from designing crop recommendation logic to integrating AI advisory and deployment, is properly verified and validated. Since the system deals with real-time agricultural decisions like irrigation alerts and fertilizer usage, accuracy and reliability are very important. The V-Model helps in detecting errors early and ensures that the final system meets the expectations of farmers.

Basic V-Model Cycle Diagram

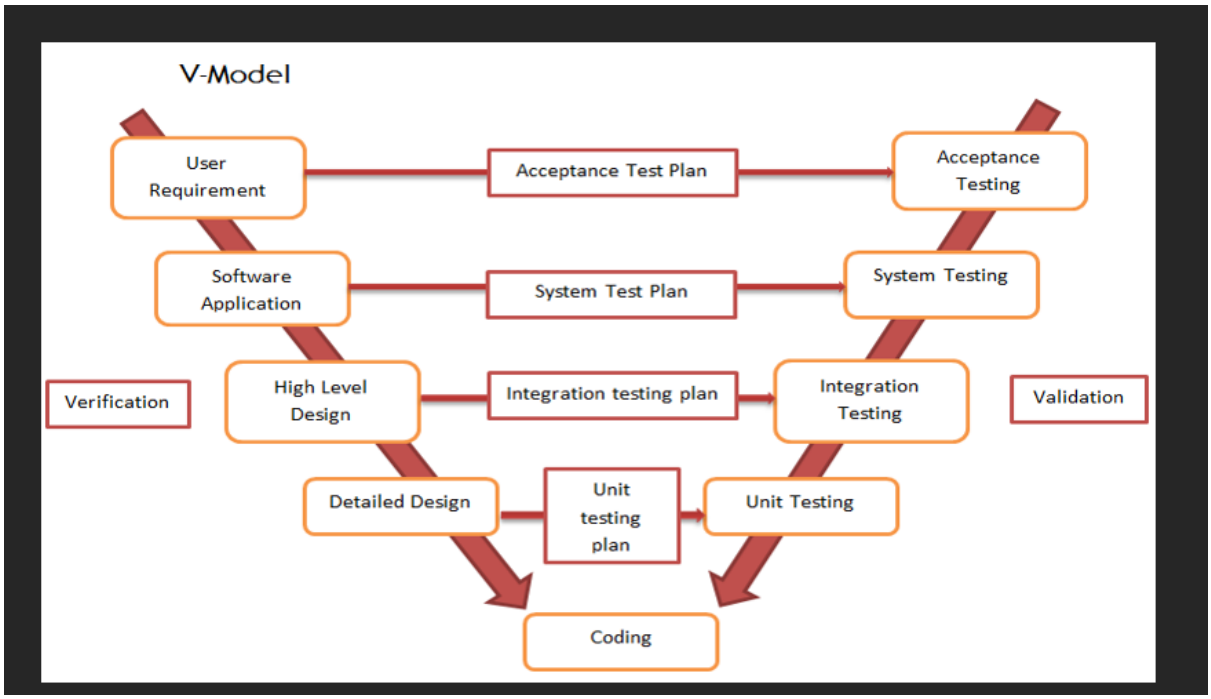

Fig 3.1 V-Model Methodology

The V-Model diagram for the Farmekox system represents a systematic development process where each development stage is paired with a corresponding testing stage. The left side of the

diagram shows the verification phases, while the right side shows validation phases, with implementation at the bottom acting as the connecting point.

Verification Phase (Left Side)

The verification phase focuses on building the system correctly according to the defined requirements.

• Requirements Analysis:

At this stage, the requirements of the system are identified and clearly defined. These include features such as crop recommendation, weather forecasting, irrigation alerts, fertilizer calculator, AI advisory (voice and chat), and market price updates. This step ensures that the system objectives are clearly understood before development begins.

• System Design:

In this phase, the overall architecture of the Farmekox system is planned. It includes designing the frontend (React.js interface), backend (Node.js and Firebase), AI integration (Gemini API), and external services like Google Maps for nearby resources. The interaction between these components is also defined.

• Module Design:

The system is divided into smaller modules such as crop recommendation module, weather alert module, fertilizer calculator, AI advisory, and market watch. Each module is designed to perform a specific function, making the system easier to develop and manage.

• Implementation (Coding):

This is the stage where the actual development takes place. The frontend is developed using React.js and Tailwind CSS, while the backend logic is implemented using Node.js and Firebase services. AI functionalities are integrated using Genkit and Google Gemini API to provide intelligent advisory.

Central Phase (Bottom of V)

• AI Advisory and Data Processing Core:

This is the core part of the Farmekox system. It involves processing user inputs such as location, crop queries, and fertilizer details. The system combines rule-based logic with AI models to generate accurate recommendations. This stage acts as a bridge between development and testing.

Validation Phase (Right Side)

The validation phase ensures that the developed system works correctly and meets user expectations.

• Unit Testing:

Each module, such as crop recommendation, weather alerts, and fertilizer calculator, is tested individually to verify its functionality and correctness.

• Integration Testing:

Different modules are tested together to ensure proper communication between frontend, backend, and AI services. For example, checking whether user input from the frontend correctly reaches the backend and returns valid results.

• System Testing:

The complete system is tested as a whole to ensure that all components work together smoothly. This includes checking performance, usability, and response time.

• User Acceptance Testing:

The system is tested from the farmer's perspective to ensure ease of use, accuracy of recommendations, and proper functioning of features like voice advisory and local language support.

• Deployment & Monitoring:

Finally, the system is deployed as a web application using Firebase hosting. Continuous monitoring is done to ensure the system performs well and handles real-time user requests effectively.

Key Features of the V-Model

1. The V-Model establishes a clear relationship between development and testing phases.
2. Each stage is validated with a corresponding testing activity, reducing errors early.
3. Requirements are linked with acceptance testing, ensuring user satisfaction.
4. System design is linked with system testing, ensuring overall functionality.
5. Module design is linked with unit testing, ensuring correctness of individual components.
6. This structured approach improves the reliability and quality of the Farmekox system.

Overall, the V-Model provides a disciplined and reliable framework for developing the Farmekox system, ensuring that both functionality and accuracy are maintained throughout the project lifecycle.

3.2 Requirements Specification

The **Requirements Specification** phase is an important step in the V-Model, where all the functional and non-functional requirements of the system are clearly defined before development begins. In this phase, the needs of end users (farmers) are carefully analyzed, and system behavior is documented in a structured way. This helps in avoiding confusion during later stages and ensures that the developed system meets real-world expectations.

For the *Farmekox – Smart Crop Advisory System*, the requirements were identified based on common challenges faced by farmers such as lack of timely information, difficulty in accessing expert advice, and absence of integrated platforms. From our understanding and implementation, we focused on building a system that is simple, informative, and accessible even to users with minimal technical knowledge.

• **Reliability:**

The system should provide accurate and consistent results, especially for crop recommendations and alerts.

• **Security**:

User data should be protected using authentication and secure communication mechanisms.

• **Availability:**

The system should be accessible anytime through web browsers on mobile and desktop devices.

3**. Input Requirements**

The system accepts various inputs from users and external sources:

• User loation (manual input or GPS-based)

• Crop-related queries

• Season and environmental conditions

• Fertilizer barcode (via camera scan)

• User preferences and language selection

4. Output Requirements

The system generates outputs in a clear and understandable format:

• Recommended crops based on conditions

• Crop calendar schedules

• Weather forecasts and irrigation alerts

• Fertilizer information and dosage suggestions

• AI-generated advisory responses (text and voice)

• Market prices and nearby resource details

5. System Constraints

• The system depends on internet connectivity for real-time data such as weather and market prices.

• Accuracy of recommendations depends on quality of input data provided by users.

• External APIs (such as weather and maps) may have rate limits or delays.

3.3 System and Functional Design

The Farmekox system is designed as a modular and scalable smart agricultural platform that integrates multiple components to provide accurate and real-time advisory to farmers. The system combines frontend interaction, backend processing, AI intelligence, and external data sources to deliver efficient and user-friendly services. Each module is responsible for a specific task, ensuring better performance, maintainability, and scalability of the overall system.

3.3.1 System Architecture

The architecture of the Farmekox system consists of interconnected layers including the user interface, backend server, AI module, database, and external APIs. These components work together to process farmer inputs and generate meaningful outputs such as crop recommendations, weather alerts, and advisory responses.

3.3.2 Web Application (Frontend):

The frontend is developed using React.js and Tailwind CSS, which acts as the main interface between the user and the system. It allows farmers to interact with features such as crop recommendation, crop calendar, AI advisory, and market watch. The interface is designed to be simple and easy to use so that farmers with limited technical knowledge can navigate without difficulty. It also supports multilingual interaction, enabling users to access the system in Kannada and English.

3.3.3 Backend Server:

The backend is implemented using Node.js and Firebase services, which handle the application logic and data processing. It receives requests from the frontend, processes user inputs, and communicates with AI modules and external APIs. The backend ensures smooth data flow, quick response time, and efficient handling of multiple user requests.

3.3.4 AI Advisory Module:

The AI module is integrated using Genkit and Google Gemini API, which provides intelligent recommendations and responses based on user queries. It analyzes inputs such as crop type,

weather conditions, and farming issues to generate personalized advisory. This module supports both text and voice interaction, making it more accessible for farmers.

3.3.5 External Services Integration:

The system integrates external APIs such as weather services and Google Maps. Weather APIs provide real-time forecasts and rainfall alerts, while Google Maps helps in identifying nearby resources like fertilizer shops, seed stores, and markets. These services enhance the accuracy and usefulness of the system.

3.3.6 Database and Authentication:

MongoDB is used for storing application data such as crop information, user details, and advisory records. Firebase Authentication is used to manage secure user login and access control. This ensures data safety and improves system reliability.

Functional Design

The functional design explains how the Farmekox system operates step-by-step to deliver agricultural advisory services effectively.

3.3.7 Workflow:

• The user opens the Farmekox web application on a mobile or desktop device.

• The user accesses the dashboard and selects a required feature such as crop recommendation, weather alerts, or AI advisory.

• The user provides inputs such as location, crop query, or fertilizer barcode.

• The frontend sends the input data to the backend server through API requests.

• The backend processes the request using AI models, database information, and external APIs.

• The AI module analyzes the data and generates suitable recommendations or responses.

• The processed results are returned to the frontend.

• The system displays outputs such as crop suggestions, weather updates, fertilizer details, or advisory messages.

The Farmekox system is designed to provide a smooth and efficient user experience by integrating multiple modules into a single platform. When a farmer interacts with the system, the inputs are processed through backend services and AI models to generate accurate and useful outputs. The system ensures minimal effort from the user by automating analysis and presenting clear results. This structured workflow helps in delivering real-time advisory, improving decision-making, and supporting better agricultural practices.

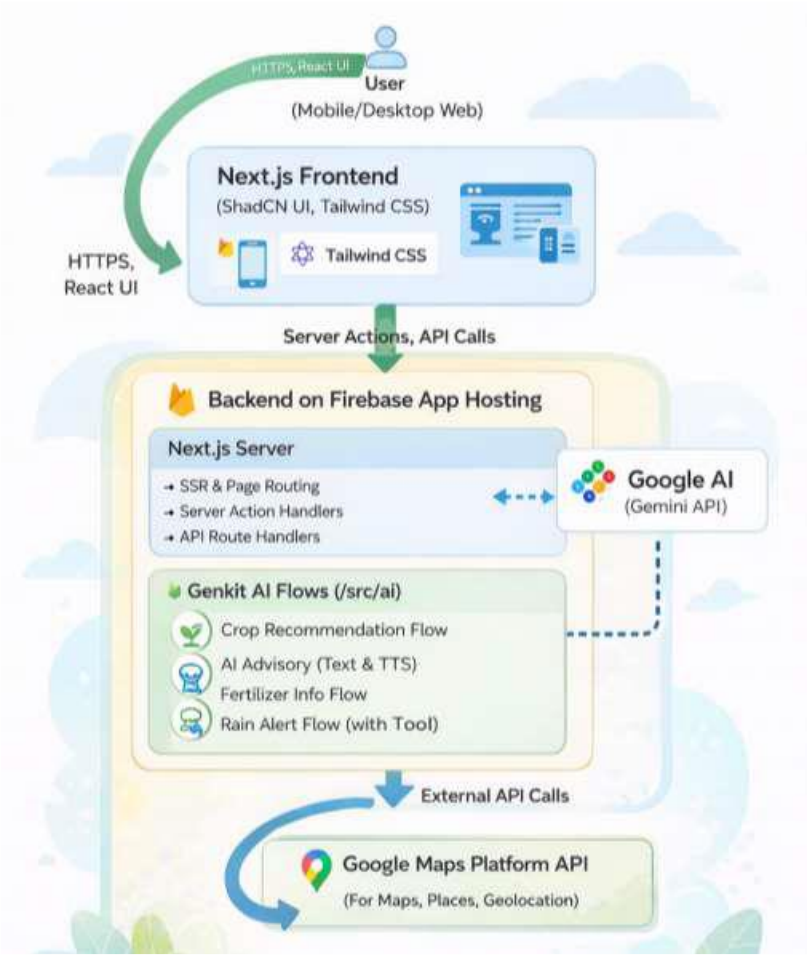

Fig 3.3 Block diagram

3.4 Unit Design and Implementation

The Farmekox system is divided into smaller functional modules to ensure efficient development, easy testing, and better maintenance. Each module is designed to perform a specific task independently while interacting smoothly with other modules in the system. This

modular approach improves system reliability, simplifies debugging, and allows future enhancements without affecting the entire application.

3.4.1 Crop Recommendation Module:

This module provides suitable crop suggestions based on user inputs such as location, season, and climatic conditions. It analyzes environmental factors and helps farmers choose crops that are best suited for their region, improving productivity and reducing risk.

3.4.2 Crop Calendar Module:

This module provides a structured schedule for farming activities such as sowing, irrigation, fertilization, and harvesting. It helps farmers follow a proper timeline and manage their agricultural activities efficiently.

3.4.3 Weather and Irrigation Alerts Module:

This module retrieves real-time weather data using external APIs and provides alerts related to rainfall and temperature. It suggests actions such as skipping irrigation during rainfall, helping farmers conserve water and optimize resource usage.

3.4.4 Fertilizer Calculator Module:

This module allows farmers to scan fertilizer barcodes using their mobile device. It provides details such as nutrient composition and recommended dosage, ensuring proper fertilizer usage and reducing soil damage.

3.4.5 Market Watch Module:

This module provides real-time mandi prices and agricultural news. It helps farmers stay updated with market trends and make informed decisions about selling their produce.

3.4.6 AI Advisory Module:

The AI module is integrated using Genkit and Google Gemini API to provide intelligent advisory. It supports both text and voice interaction in multiple languages like Kannada and English, allowing farmers to ask queries and receive personalized guidance.

3.4.7 Ask Expert Module:

This module enables farmers to connect with agricultural experts for personalized advice. It helps in solving specific problems that may not be addressed by automated systems.

3.4.8 Nearby Resources Module:

This module uses location services to identify nearby agricultural resources such as seed stores, fertilizer shops, and markets. It integrates with mapping services to provide accurate directions and location details.

3.4.9 Direct Sales Module:

This module allows farmers to directly connect with buyers and sell their products without intermediaries. It helps in improving farmer income and reducing dependency on middlemen.

3.4.10 Implementation Approach

The Farmekox system is implemented using a modular programming approach where each module is developed and tested independently. This makes the system easier to maintain and update. Proper interfaces are defined to ensure smooth communication between modules. This design also supports scalability, allowing new features to be added without affecting existing functionalities.

Chapter 4

Project Management

4.1 Project Timeline

A project timeline is used to represent the sequence of tasks and activities required to complete the project. It is commonly represented using a Gantt chart, which provides a visual representation of tasks, duration, and dependencies. The timeline helps in understanding how the project progresses from initial planning to final deployment.

A Gantt chart includes tasks, duration, start and end dates, dependencies between tasks, and milestones. It helps in planning, scheduling, tracking progress, and ensuring that all activities are completed within the deadline. In the Farmekox project, the development was divided into phases such as requirement analysis, design, development, testing, and deployment.

Table 4.1 Project Planning Timeline

Phase	Activity	Duration	Description
Phase1	Requirement Analysis	1 Week	Understanding farmer needs and defining system requirements
Phase 2	Literature Review	1 Week	Studying existing systems and identifying gaps
Phase 3	System Design	2 Weeks	Designing architecture and modules
Phase 4	Module Development	4 Weeks	Developing core modules like crop recommendation, AI advisory
Phase 5	Integration	2 Weeks	Integrating frontend, backend, and AI components
Phase 6	Testing	2 Weeks	Performing unit, integration, and system testing
Phase 7	Deployment	1 Week	Deploying the application and final review

Table 4.1 shows the timeline during the planning phase.

Table 4.2 Project Implementation Timeline

Module	Activity	Duration	Description
Crop Recommendation	Development & Testing	1 Week	Implementing crop suggestion logic
Weather Module	API Integration	1 Week	Integrating weather data and alerts
Fertilizer Module	Barcode Implementation	1 Week	Developing fertilizer calculator
AI Advisory	AI Integration	2 Weeks	Implementing Gemini API for advisory
Market Watch	Data Integration	1 Week	Fetching mandi price data
UI Development	Frontend Design	2 Weeks	Designing user interface
Final Testing	System Testing	1 Week	Testing complete system

Table 4.2 implementation phase timeline

It showing how each module was developed and tested individually before integration.

The timeline followed in this project helped in maintaining a structured workflow and ensured that all modules were completed within the planned duration.

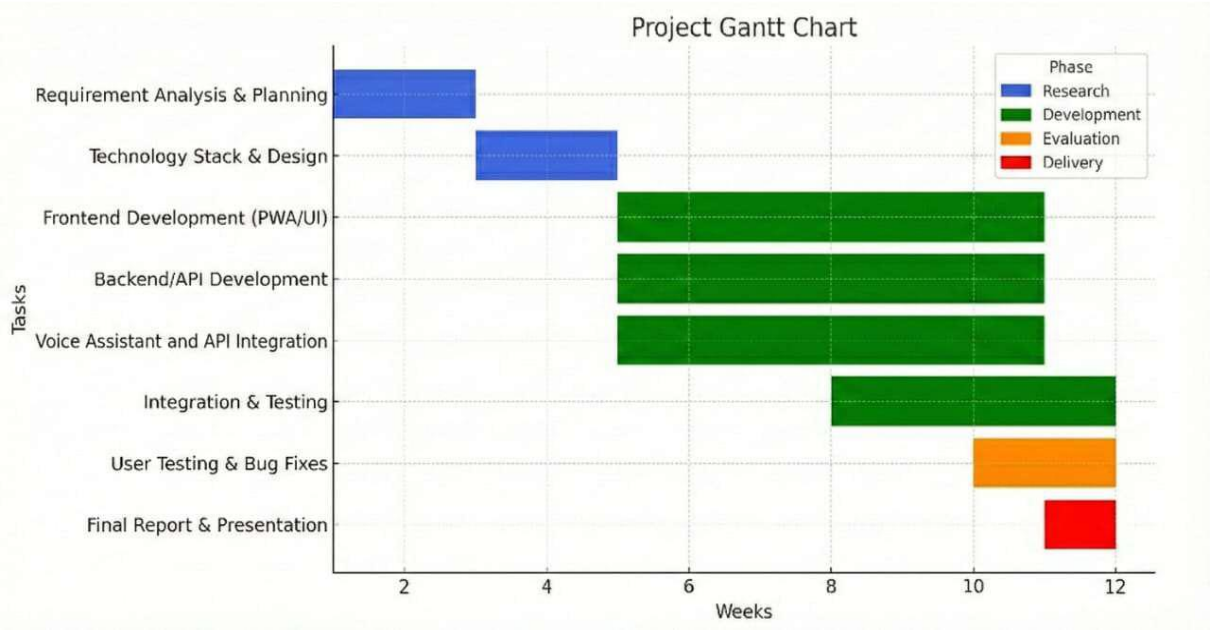

Fig 4.1: Gantt Chart

4.2 Risk Analysis

Risk analysis is an important part of project management as it helps in identifying potential risks and planning mitigation strategies. In this project, a **PESTLE analysis** approach is used to evaluate different types of risks.

Table 4.3 PESTLE Analysis

Factor	Risk	Impact	Mitigation
Political	Changes in agricultural policies	Moderate	Design system adaptable to policy updates
Economic	Cost of internet access	Moderate	Optimize system for low data usage
Social	Low digital literacy among farmers	High	Provide simple UI and local language support
Technological	API failures or downtime	High	Use fallback mechanisms and caching
Legal	Data privacy issues	High	Implement secure authentication and data protection
Environmental	Unpredictable weather conditions	High	Use real-time weather APIs

Table 4.3 shows different risk factors affecting the project and corresponding mitigation strategies.

Project Risk Matrix:

High risks such as API failures and data privacy are handled through secure design and backup mechanisms. Moderate risks such as economic and policy changes are managed through flexible system design.

Overall, risk analysis helps in identifying challenges early and ensures smooth project execution.

4.3 Project Budget

Project budgeting involves estimating the cost required for developing and deploying the system. The Farmekox project mainly uses open-source technologies, which helps in reducing

development cost.

Table 4.4 Project Budget

Resource	**Description**	**Cost (₹)**
Development Tools	VS Code, GitHub (Free)	0
Cloud Services	Firebase (Free Tier)	0
API Services	Weather API, Maps API	0–1000
Internet Charges	Development and testing	1000
Miscellaneous	Documentation and testing	500
Total Estimated Cost		~1500–2500

Table 4.4 Cost of the project.

Since most tools used are free or open-source, the overall cost is minimal.

The budgeting process involved identifying resources, estimating costs, and ensuring efficient utilization. From our implementation, we observed that using cloud-based and open-source tools significantly reduces project cost while maintaining performance.

Overall, project management ensured that the Farmekox system was developed in a structured and efficient manner. Proper planning, risk management, and budgeting contributed to the successful completion of the project within the given time and resources.

Chapter 5

Analysis and Design

Analysis and Design

Analysis and design are two closely related phases in system development. Analysis focuses on understanding the real-world problem by breaking it into smaller parts and identifying requirements. Design focuses on creating a structured solution that satisfies those requirements. In the case of the *Farmekox – Smart Crop Advisory System*, the analysis phase involved studying the challenges faced by farmers such as dependency on traditional knowledge, lack of timely information, and absence of a unified platform. Based on this, the design phase focuses on building a scalable, modular, and intelligent web-based system.

From our implementation, we observed that combining multiple agricultural services into a single platform improves usability and efficiency. The design ensures that each module works independently but is well-integrated with the system. The system is designed using a layered architecture where frontend, backend, AI, and database components interact seamlessly.

5.1 Requiretments

The requirements define what the system should do and how it should perform. These are categorized into **Hardware (HW)** and **Software (SW)** requirements along with additional functional requirements.

System Hardware (HW) Requirement Phase

1.Identify Initial Conditions:

The system operates in a digital environment where users access it through smartphones or computers. It assumes that users have basic internet connectivity. No physical sensors are directly used, but user devices act as input sources.

2.Determine Input Parameters:

Inputs include user location (manual or GPS), crop selection, seasonal data, fertilizer barcode scan, and queries for AI advisory. External inputs include weather data and market prices retrieved through APIs.

3.System Outcomes:

The system produces outputs such as crop recommendations, crop calendars, irrigation alerts, fertilizer dosage suggestions, AI advisory responses, and mandi prices.

4.Formulate Relations:

User Input → Frontend → Backend → AI Processing → Database/API → Output

This relationship ensures smooth flow of information.

5.Identify System Constraints:

The system depends on internet availability, API reliability, and accuracy of user input. Performance may be affected by server load and API latency.

System Software (SW) Requirement Phase

1. Identify Initial Conditions:

The system runs on web browsers using a client-server architecture.

2. Determine Input Parameters:

Inputs include user data, API responses, and stored database data.

3. System Outcomes:

Outputs include processed data such as recommendations, alerts, and advisory messages.

4.Formulate Relations:

Frontend → API Calls → Backend → AI Model → Database → Response

5.Identify System Constraints:

Constraints include handling multiple users, API limitations, and system scalability.

Additional Requirements

• Data Collection Requirements:

Data is collected from user inputs and external sources such as weather APIs and market databases.

• **Data Analysis Requirements:**

Data is processed using AI models (Genkit + Gemini API) and rule-based logic.

• **System Management Requirements:**

The system should provide a centralized dashboard for easy navigation.

• **Security Requirements:**

Authentication and secure data transmission must be ensured.

• **User Interface Requirements:**

The interface should be simple, responsive, and multilingual to support farmers.

Table 5.1 Summarizing Requirements

Requirement	Description
Purpose	Smart crop advisory system for farmers
Behaviour	Provides real-time recommendations
System Management	Centralized dashboard
Data Analysis	AI-based processing
Application Deployment	Web-based system
Security	Secure authentication

System Hardware Design Phase

1. Identification of functional blocks (Input, Processing, Output)
2. Development of system workflow
3. Identification of inconsistencies in data flow
4. Designing interaction interfaces
5. System-level design and analysis
6. Development of integrated testing strategy

System Software Design Phase

1. Identification of components (React frontend, Node backend, AI module)
2. Development of application workflow
3. Identification of possible system errors
4. Designing REST APIs
5. System design and performance evaluation
6. Development of testing framework

5.2 Block diagram

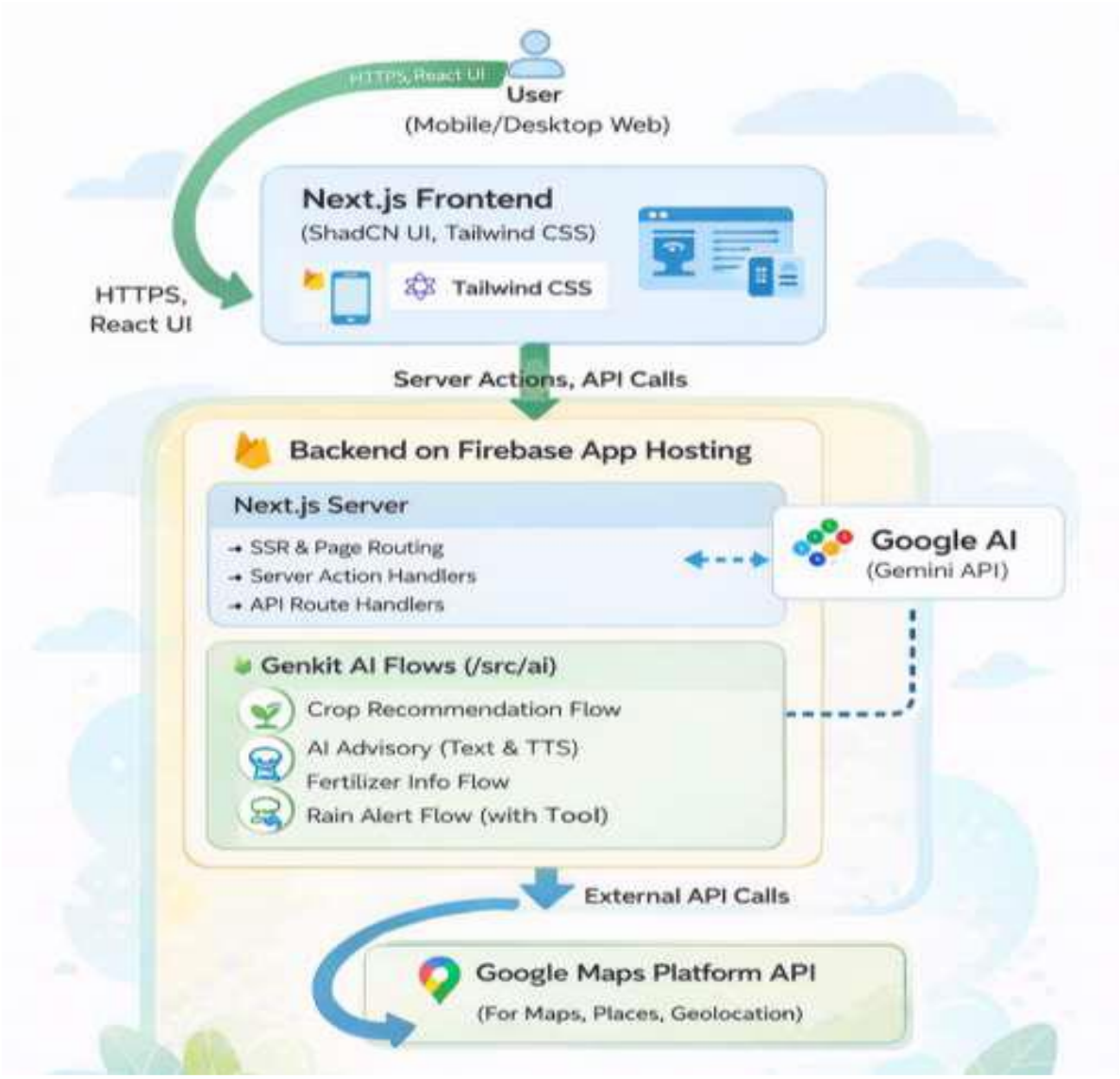

Fig 5.1 Functional Block Diagram

The functional block diagram of the *Farmekox – Smart Crop Advisory System* is shown in Fig

5.1. The system is designed using a layered architecture consisting of input, processing, and output blocks. The diagram clearly represents how user inputs are processed through different system components to generate meaningful outputs.

The **input block** consists of the user accessing the system through mobile or desktop web browsers. The user interacts with the system using a web interface built using Next.js frontend along with ShadCN UI and Tailwind CSS. The communication between the user and the system takes place through HTTPS requests.

The **processing block** is the core part of the system. It includes the backend hosted on Firebase App Hosting, which runs the Next.js server. This server handles server-side rendering (SSR), page routing, API route handling, and server actions. The backend processes user requests and communicates with AI services and external APIs.
Within the processing block, the system uses **Genkit AI flows**, which include multiple modules such as crop recommendation flow, AI advisory flow (text and text-to-speech), fertilizer information flow, and rain alert flow. These flows are responsible for analyzing user input and generating intelligent responses.

The system integrates with **Google AI (Gemini API)**, which provides advanced AI capabilities for generating recommendations and advisory responses. This improves the accuracy and intelligence of the system.
The system also interacts with external services through API calls. The **Google Maps Platform API** is used for location-based services such as maps, geolocation, and nearby resource identification.

The **output block** consists of the results generated by the system, which are displayed to the user through the frontend interface. These outputs include crop recommendations, advisory responses, weather alerts, fertilizer suggestions, and location-based services.
The arrows in the diagram indicate the flow of data between different components. The user sends requests through the frontend, which are processed by the backend and AI modules, and the final output is returned to the user.
From the implementation, it is observed that this block diagram ensures a clear separation of

concerns between frontend, backend, AI processing, and external services. This improves scalability, maintainability, and system performance.

5.3 System Flow chart

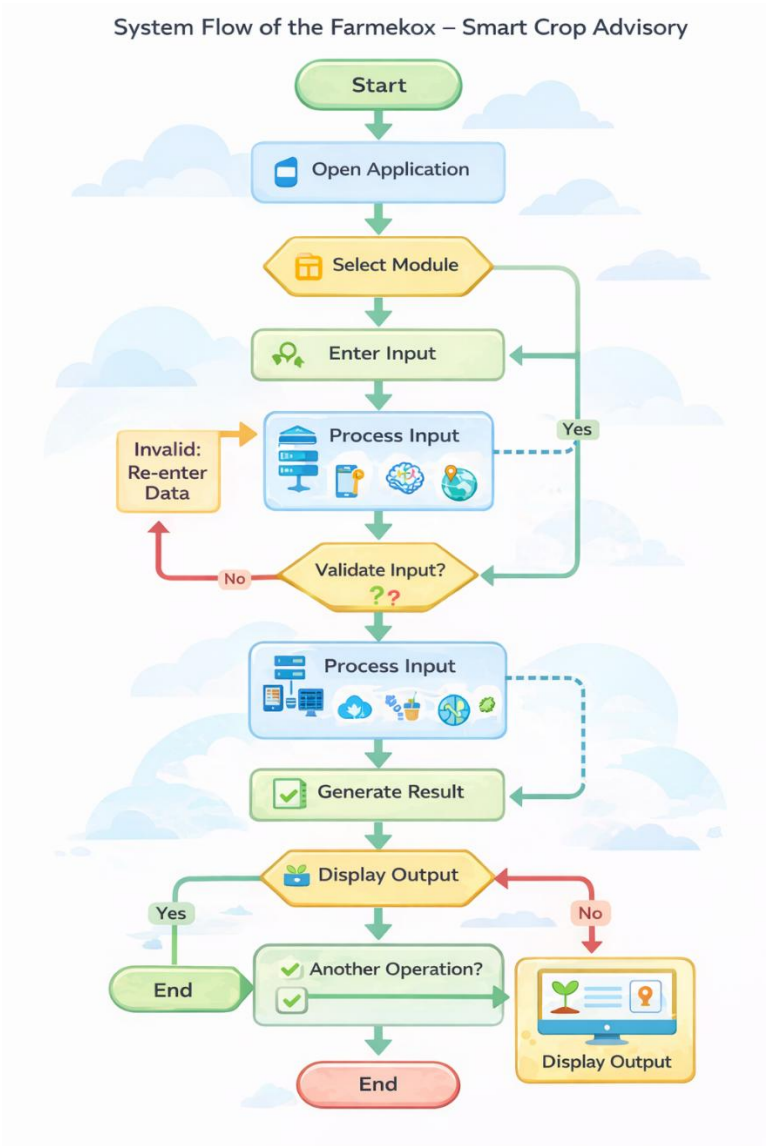

Fig 5.3 System Flow Chart

The system flow chart of the *Farmekox – Smart Crop Advisory System* is shown in Fig 5.2. The flowchart represents the step-by-step working of the system from user interaction to output generation. It helps in understanding the logical sequence of operations performed by the system.

The flow begins with the **initialization stage**, where the user opens the application using a mobile or desktop web browser. The system loads the frontend interface and establishes

communication with the backend server through secure HTTPS protocols.

In the next step, the **user selects a module** such as crop recommendation, AI advisory, fertilizer information, or weather alerts. Based on the selected module, the system prompts the user to enter relevant inputs such as location, crop type, or query.

After receiving the input, the system performs **input validation** to ensure that the provided data is correct and complete. If the input is invalid, the system prompts the user to re-enter the data. If the input is valid, the data is forwarded to the backend server for processing.

The backend then performs **data processing and API communication**. It interacts with AI modules and external APIs such as weather services and maps. The system uses Genkit AI flows and Gemini API to analyze the input and generate intelligent responses.

Once processing is completed, the system generates **output results** such as crop recommendations, advisory messages, fertilizer suggestions, or alerts. These results are then sent back to the frontend.

Finally, the **output is displayed to the user** through the web interface. The user can either view the results or perform another operation by selecting a different module, thereby continuing the flow.

From the implementation, it is observed that the system flow ensures smooth interaction between user input, backend processing, and output generation. The use of validation and modular processing improves system reliability and efficiency.

5.4 Hardware Components and Specifications

The Farmekox system is a web-based application and does not require dedicated IoT hardware components such as sensors or microcontrollers. Instead, it relies on general-purpose devices such as smartphones, laptops, and desktops to access the system. These devices act as input, processing, and output units.

The system is designed to be compatible with devices that have basic computing capabilities and internet connectivity. Users interact with the system through a web browser, which eliminates the need for specialized hardware. This approach reduces the cost of deployment and makes the system easily accessible to a wide range of users, especially farmers in rural areas.

The performance of the system depends on the device specifications such as processor speed,

RAM, and internet connectivity. Smartphones are the most commonly used devices among farmers due to their portability and affordability. Therefore, the system is developed using a mobile-first approach to ensure smooth performance even on low-end devices.

The system requires stable internet connectivity to access cloud services, APIs, and real-time data such as weather updates and market prices. Any interruption in connectivity may affect system performance and data accuracy.

There are certain constraints related to hardware usage. Low-end devices may experience slower response time, and mobile devices may face battery limitations during continuous usage. However, the system is optimized to minimize resource usage and provide efficient performance.

Overall, the use of general-purpose hardware makes the Farmekox system flexible, cost-effective, and scalable. It ensures that the application can be used widely without requiring additional investment in specialized hardware components.

5.5 Software Components and Requirements

The Farmekox system is developed using modern web technologies and cloud-based services. The software components are selected based on performance, scalability, and ease of development. These components work together to provide a complete smart crop advisory system.

Component	**Specification**
Frontend Development	React.js, Next.js
UI Design	Tailwind CSS, ShadCN UI
Backend	Node.js
Cloud Platform	Firebase (Hosting, Authentication)
Database	MongoDB
AI Integration	Genkit, Google Gemini API
External APIs	Weather API, Google Maps API

Component	Specification
Programming Language	JavaScript, TypeScript
Build Tool	npm, Vite
Version Control	Git, GitHub

Table 5.5 Components and Requirements

The frontend is developed using React.js and Next.js, which provide a dynamic and responsive user interface. Tailwind CSS and ShadCN UI are used for designing clean and user-friendly layouts. The backend is implemented using Node.js, which handles server-side logic and API communication.

Firebase is used for hosting and authentication, ensuring secure and scalable deployment. MongoDB is used as the database for storing application data. AI functionality is implemented using Genkit and Google Gemini API to provide intelligent advisory responses.

External APIs such as weather services and Google Maps are integrated to provide real-time data and location-based features. The system uses JavaScript and TypeScript as programming languages, and tools like npm and Vite are used for building the application. Version control is managed using Git and GitHub.

5.6 Functional Modules

Functional Modules The Farmekox system is divided into multiple functional modules, where each module performs a specific task. This modular approach helps in improving system efficiency, scalability, and maintainability. Each module is designed independently but works together as part of the complete system.

Table 5.6 Functional Modules Description

Module	Description
Crop Recommendation	Suggests suitable crops based on location, season, and environmental conditions
Crop Calendar	Provides a schedule for farming activities such as sowing, irrigation, and harvesting

Module	Description
Weather & Irrigation Alerts	Gives real-time weather updates and irrigation recommendations
Fertilizer Calculator	Provides fertilizer details and dosage using barcode scanning
AI Advisory	Offers intelligent advice using chat and voice in Kannada and English
Market Watch	Displays current mandi prices and agricultural market trends
Ask Expert	Allows users to get personalized advice from experts
Nearby Resources	Shows nearby agricultural shops, markets, and services using location
Direct Sales	Helps farmers connect directly with buyers to sell their produce

Each module in the Farmekox system is designed to solve a specific problem faced by farmers. The crop recommendation module helps in selecting suitable crops, while the crop calendar guides farmers throughout the farming cycle. Weather and irrigation alerts assist in efficient water management.

The fertilizer calculator reduces the chances of incorrect fertilizer usage by providing accurate dosage information. The AI advisory module plays a key role by providing intelligent suggestions based on user queries in multiple languages, making the system more accessible.

The market watch module helps farmers make better selling decisions by providing real-time price information. The ask expert feature allows farmers to seek guidance from professionals. Nearby resources help users find essential services, and the direct sales module supports farmers in connecting with buyers.

From the implementation, it is observed that dividing the system into modules improves performance, simplifies development, and makes the system more user-friendly.

5.7 Datasets and data resources

Datasets and data resources [8] The Farmekox system is divided into multiple functional modules, where each module performs a specific task. This modular approach helps in improving system efficiency, scalability, and maintainability. Each module is designed independently but works together as part of the complete system.

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Chapter 6

Hardware, Software and Simulation

6.1 Hardware

Hardware, Software and Simulation

This chapter describes the hardware and software components used in the development of the Farmekox system. It also explains the implementation process and simulation aspects. The system is primarily software-based, so the focus is on development tools, coding, and system behavior.

The Farmekox system does not use dedicated IoT hardware components such as sensors, actuators, or microcontrollers. Instead, it relies on general-purpose devices such as smartphones, laptops, and desktops for accessing the application.

These devices act as input, processing, and output units. The user interacts with the system through a web browser, which eliminates the need for specialized hardware. This reduces cost and improves accessibility for farmers.

During development, laptops and desktops are used due to their higher processing power and ease of coding. Smartphones are mainly used for testing the responsiveness and usability of the application.The system requires stable internet connectivity to access backend services, APIs, and cloud databases. The hardware configuration should support modern web browsers such as Chrome, Edge, or Firefox. Since no physical hardware integration is involved, there is no need for circuit design or interfacing components. However, the system is designed to be compatible with future integration of IoT devices if required.

6.2 Software Development Tools

The Farmekox system uses various software development tools to design, develop, test, and deploy the application. These tools help in improving development efficiency and maintaining code quality. The frontend of the application is developed using React.js and Next.js, which provide a dynamic and responsive user interface. Tailwind CSS is used for styling, which simplifies UI design and ensures consistency.

The backend is developed using Node.js, which handles server-side logic, API communication, and data processing. Firebase is used for hosting and authentication, providing secure and scalable cloud services. MongoDB is used as the database to store application data such as user

details, crop information, and advisory records. It provides flexibility in handling large and unstructured data.

Version control is managed using Git and GitHub, which helps in tracking changes and collaboration during development. API testing is performed using tools like Postman to ensure proper functionality of backend services.

The development environment includes code editors such as Visual Studio Code, which provides features like debugging, extensions, and syntax highlighting. The application is built and managed using npm and other package managers.

These tools together provide a complete development ecosystem for building and maintaining the Farmekox system.

6.3 Software Code

The software of the Farmekox system is developed using JavaScript and TypeScript. The code is divided into frontend, backend, and AI modules. The following sample code represents the working of the crop recommendation and API handling module.

Frontend Code (React Component)

The frontend collects user input and sends it to the backend.

```
import { useState } from "react";

function CropForm() {
  const [location, setLocation] = useState("");
  const [result, setResult] = useState("");

  const handleSubmit = async () => {
    // Sending user input to backend API
    const response = await fetch("/api/recommend", {
      method: "POST",
      headers: {
        "Content-Type": "application/json"
      },
      body: JSON.stringify({ location })
    });
```

```
    const data = await response.json();
    setResult(data.crop);
  };

  return (
    <div>
      <input
        type="text"
        placeholder="Enter Location"
        onChange={(e) => setLocation(e.target.value)}
      />
      <button onClick={handleSubmit}>Get Recommendation</button>
      <p>{result}</p>
   </div>
  );
}
export default CropForm;
```

Backend Code (Node.js API)

The backend processes user input and generates a response.

```
export default async function handler(req, res) {
  if (req.method === "POST") {
    const { location } = req.body;

    // Simple logic for crop recommendation
    let crop;

    if (location.toLowerCase().includes("karnataka")) {
      crop = "Rice, Ragi";
    } else {
      crop = "Wheat, Maize";
```

```
    }
// Sending response back to frontend
    res.status(200).json({ crop });
  } else {
    res.status(405).json({ message: "Method Not Allowed" });
  }
}
```

AI Integration Code (Gemini API – Concept)

The AI module processes user queries and generates intelligent responses.

```
import { GoogleGenerativeAI } from "@google/generative-ai";

const genAI = new GoogleGenerativeAI("YOUR_API_KEY");

export async function getAdvisory(query) {
  // Initialize model
  const model = genAI.getGenerativeModel({ model: "gemini-pro" });

  // Generate response
  const result = await model.generateContent(query);
  const response = await result.response;

  return response.text();
}
```

The frontend code is responsible for collecting user input and displaying results. It sends requests to the backend using API calls. The backend processes the input and generates appropriate responses based on logic or data.

The AI module enhances the system by providing intelligent advisory responses using the Gemini API. This allows the system to understand natural language queries and generate useful suggestions. From the implementation, it is observed that modular coding improves readability.

Chapter 7

Results and Evaluation

7.1 Test points

Evaluation and Results

This chapter presents the evaluation of the Farmekox system through various testing methods. The system is tested under different scenarios to ensure that it meets the objectives and performs efficiently. The evaluation includes identifying test points, developing test plans, recording results, and analyzing system performance.

Test points are specific locations or stages in the system where testing is performed to verify functionality. In the Farmekox system, test points are identified in different modules such as frontend, backend, AI advisory, and API integration.

The following test points are considered:

- **TP1:** User input validation (location, crop query)
- **TP2:** API request handling between frontend and backend
- **TP3:** Backend processing and logic execution
- **TP4:** AI advisory response generation
- **TP5:** Weather API data retrieval
- **TP6:** Database storage and retrieval
- **TP7:** Output display on user interface

Each test point is used to check whether the system is functioning correctly at that stage.

7.2 Test Plan

The test plan defines how the system will be tested under different conditions. It includes various test cases for validating system performance.

- **TP1:** The system should accept valid user input and reject invalid input
- **TP2:** The API should return response within acceptable time
- **TP3:** Backend should process input correctly
- **TP4:** AI module should generate meaningful advisory responses
- **TP5:** Weather API should provide accurate data
- **TP6:** Database should store and retrieve data correctly

• **TP7:** Output should be displayed correctly on UI

The testing includes:

• Unit Testing – Testing individual modules

• Integration Testing – Testing interaction between modules

• System Testing – Testing complete system

• Black Box Testing – Testing functionality without internal knowledge

• White Box Testing – Testing internal logic

7.3 Test Results

The results obtained from testing are recorded and analyzed.

Table 7.1 Test Results

Test Point	Expected Output	Actual Output	Result
TP1	Valid input accepted	Accepted	Pass
TP2	API response < 2 sec	1.5 sec	Pass
TP3	Correct processing	Correct output	Pass
TP4	Accurate AI response	Relevant advisory	Pass
TP5	Correct weather data	Accurate data	Pass
TP6	Data stored/retrieved	Successful	Pass
TP7	Output displayed	Proper display	Pass

It is observed that all modules of the system are functioning correctly. The response time is within acceptable limits, and the outputs match expected results.

The AI advisory module provides meaningful responses, although slight variations may occur depending on input queries. The system performance is stable under normal conditions.

7.4 Insights

The evaluation of the system provides the following insights:

• The system performs efficiently with minimal delay

• API integration plays a major role in system performance

• AI responses depend on input quality

• The system is scalable and can handle multiple users

Some limitations observed:

• Performance may reduce with poor internet connectivity

• API dependency may affect system availability

• AI responses may vary slightly

Improvements

Based on evaluation, the following improvements can be made:

• Optimize API calls to reduce latency

• Improve AI model accuracy

• Add offline support for basic features

• Enhance UI for better user experience

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

8.1 Social Aspects

This chapter discusses the broader impact of the Farmekox system beyond technical implementation. It focuses on how the system affects society, follows legal regulations, maintains ethical standards, supports sustainability, and ensures safety. These aspects are important to evaluate the real-world applicability and responsibility of the system.

The Farmekox system has a positive impact on society, especially in the agricultural sector. It helps farmers make better decisions by providing real-time information, crop recommendations, and expert advice. This improves productivity and reduces dependency on traditional guesswork methods.

The system also promotes digital literacy among farmers by encouraging the use of technology in agriculture. By supporting regional languages such as Kannada, the system becomes more accessible to rural users.

However, there are some challenges. Farmers who do not have access to smartphones or internet connectivity may not benefit from the system. This creates a digital divide between users who can access technology and those who cannot.

Overall, the system contributes positively to society by improving agricultural practices and empowering farmers with knowledge.

8.2 Legal Aspects

The Farmekox system must comply with data protection and privacy laws. Since the system collects user data such as location and queries, it is important to ensure that this data is handled securely.

The system follows basic legal principles such as user consent, data protection, and secure communication. Firebase authentication is used to ensure secure login and prevent unauthorized access.

There are also legal considerations related to the use of third-party APIs such as weather services and maps. These APIs must be used according to their terms and conditions.

In future, the system should comply with data protection regulations such as India's Digital Personal Data Protection Act to ensure full legal compliance.

8.3 Ethical Aspects

Ethical considerations play an important role in the development of the Farmekox system. The system should provide accurate and unbiased recommendations to farmers. Any incorrect advice may lead to financial loss for users.

The AI advisory system should be transparent and should not mislead users. It should clearly indicate that recommendations are based on available data and AI analysis.

The system should respect user privacy and should not misuse personal data. Developers must ensure that the system is designed for the benefit of society and not for exploitation.

From our implementation, we observed that ethical responsibility is important in AI-based systems, especially when they impact real-world decisions.

8.4 Sustainability Aspects

The Farmekox system supports sustainability in agriculture by promoting efficient use of resources. It helps farmers optimize water usage through irrigation alerts and provides guidance on proper fertilizer usage.

The system reduces wastage by providing accurate recommendations, which leads to better crop yield and resource management. It also promotes environmentally friendly practices by encouraging sustainable farming methods.

Since the system is software-based, it does not require additional physical resources, which reduces environmental impact. Cloud-based deployment also helps in efficient resource utilization.

Overall, the system contributes to sustainable agriculture and supports long-term environmental benefits.

8.5 Safety Aspects

Safety is an important consideration in the Farmekox system. The system ensures data security through authentication and secure communication protocols.

The system should provide reliable information to avoid incorrect decisions that may affect farmers. Safety also includes protecting the system from cyber threats such as unauthorized access and data breaches.

The use of secure APIs and cloud platforms helps in maintaining system safety. Regular updates and monitoring are required to ensure system reliability.

In future, additional safety measures such as encryption, secure backups, and monitoring tools can be implemented to enhance system security.

The Farmekox system not only provides technical solutions but also considers social, legal, ethical, sustainability, and safety aspects. These factors ensure that the system is responsible, reliable, and beneficial for real-world applications.

warns them about the danger through alerts

Chapter 9

Conclusion

9.1 Project Approach

The system design adopted by Farmekox follows a modular and layered architecture. This approach ensures that different functionalities such as crop recommendation, weather alerts, fertilizer guidance, and AI advisory work independently while being integrated into a single platform. The application is developed as a web-based system using React.js for frontend and Node.js for backend processing.

The system collects user inputs such as location, crop queries, and fertilizer details, which are then processed through backend logic and AI modules. The AI advisory is implemented using Genkit and Google Gemini API, which enables the system to generate intelligent recommendations in natural language. The system also integrates external APIs such as weather services and maps to provide real-time information.

In addition, the system prioritizes accessibility and usability by supporting multiple languages such as Kannada and English. The use of Firebase ensures secure authentication and cloud-based deployment, while MongoDB is used for efficient data storage. The overall approach ensures real-time response, scalability, and ease of use for farmers.

9.2 Results Obtained

During testing and evaluation, the Farmekox system performed effectively in providing accurate and useful outputs. The system successfully generated crop recommendations based on user inputs such as location and season. The weather module provided real-time alerts, helping users plan irrigation activities efficiently.

The AI advisory module produced meaningful responses to user queries, making the system interactive and user-friendly. The system responded within a few seconds for most operations, ensuring real-time usability. The integration of external APIs worked reliably, providing accurate weather and location-based data.

Overall, the system demonstrated stable performance, with most functionalities working correctly under normal conditions. Minor delays were observed in some cases due to API response time, but they did not significantly affect system usability.

9.3 Alignment with Project Objectives

The developed Farmekox system successfully meets the objectives defined in the initial stages of the project. The system provides real-time crop recommendations and advisory services, which aligns with the objective of improving decision-making for farmers.

The integration of AI-based advisory supports the objective of providing intelligent and data-driven solutions. The system also ensures easy access to information through a user-friendly interface, fulfilling the objective of improving accessibility.

The use of cloud-based deployment and modular design supports system scalability and management objectives. Security requirements are addressed through authentication and secure data handling. Overall, the system aligns well with all major objectives including behaviour, analysis, system management, security, and deployment.

9.4 Future Recommendations

The Farmekox system can be further improved to enhance its performance and usability. One possible improvement is the integration of IoT sensors to collect real-time data such as soil moisture, temperature, and humidity. This would improve the accuracy of recommendations.

The AI advisory module can be enhanced by training with larger datasets to provide more accurate and personalized suggestions. The system can also be expanded to support more regional languages to increase accessibility.

Development of a dedicated mobile application can improve user experience and performance. Offline features can be added to support users in areas with limited internet connectivity. Integration with government agricultural schemes and services can also improve the usefulness of the system.

Overall, these improvements can make the system more robust, efficient, and widely usable in real-world agricultural scenarios.

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Base Paper

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Appendix

The appendix section provides additional supporting information related to the development and implementation of the *Farmekox – Smart Crop Advisory System*. These details help in better understanding the technical aspects and validate the work carried out.

Appendix A: Software and Technology Details

The Farmekox system is developed using modern web technologies and cloud-based services.

• React.js and Next.js – Frontend development

• Tailwind CSS and ShadCN UI – UI design

• Node.js – Backend development

• Firebase – Hosting and authentication

• MongoDB – Database

• Genkit and Google Gemini API – AI advisory

• Weather API – Real-time weather data

• Google Maps API – Location services

These technologies are selected for scalability and performance.

Appendix B: Sample Source Code

The following code shows how backend processes user input:

```
export default function handler(req, res) {
  const { location } = req.body;
  let crop;

  if (location.toLowerCase().includes("karnataka")) {
    crop = "Rice, Ragi";
  } else {
    crop = "Wheat, Maize";
  }

  res.status(200).json({ crop });
}
```

This code demonstrates input handling and response generation.

Appendix C: System Screenshots

This section includes screenshots of:

• Home page

• Crop recommendation module

• AI advisory chat

• Weather alerts

• Market watch

These screenshots show the working interface of the system.

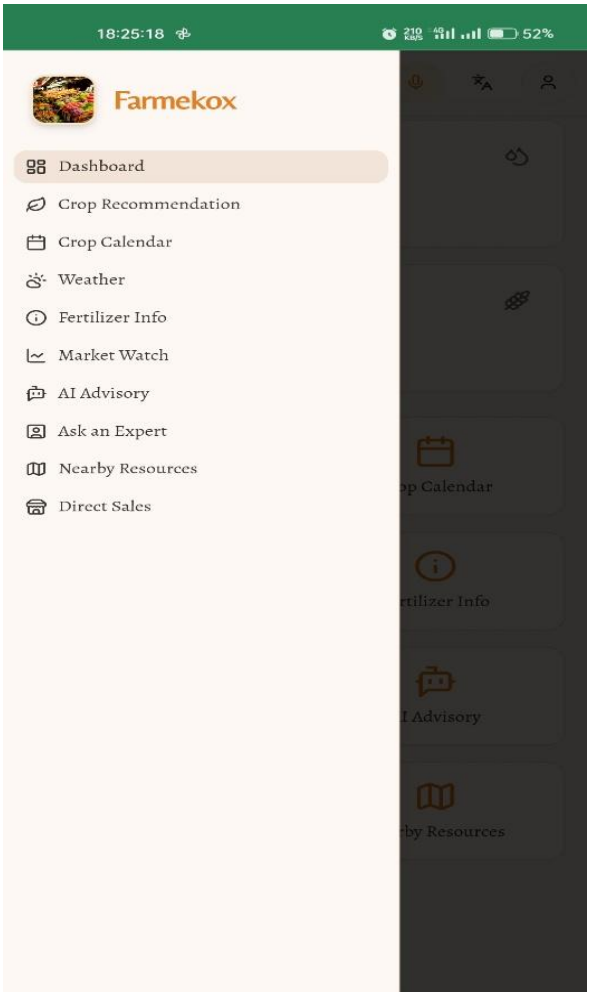

Fig 9.1: Homepage

Fig 9.2: Ai chat

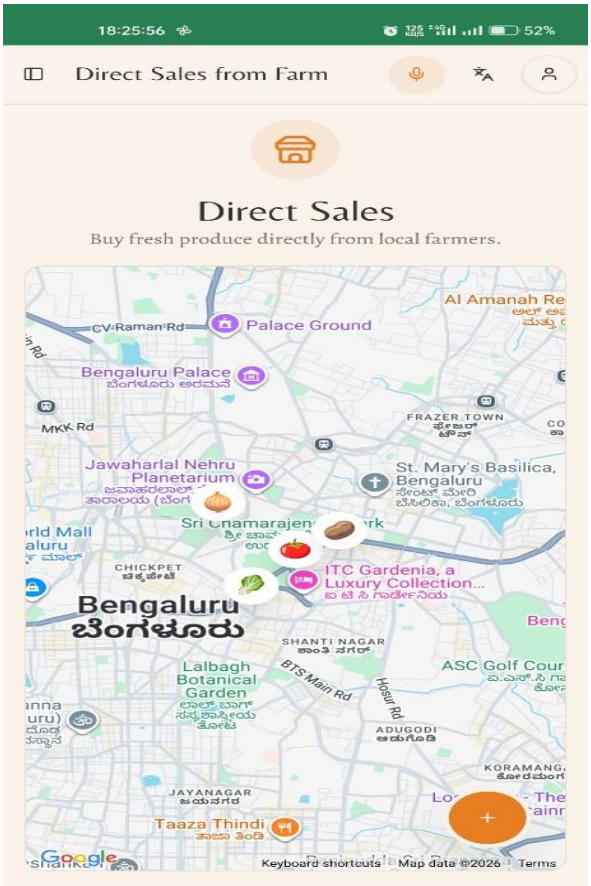

Fig 9.3: Direct Sales

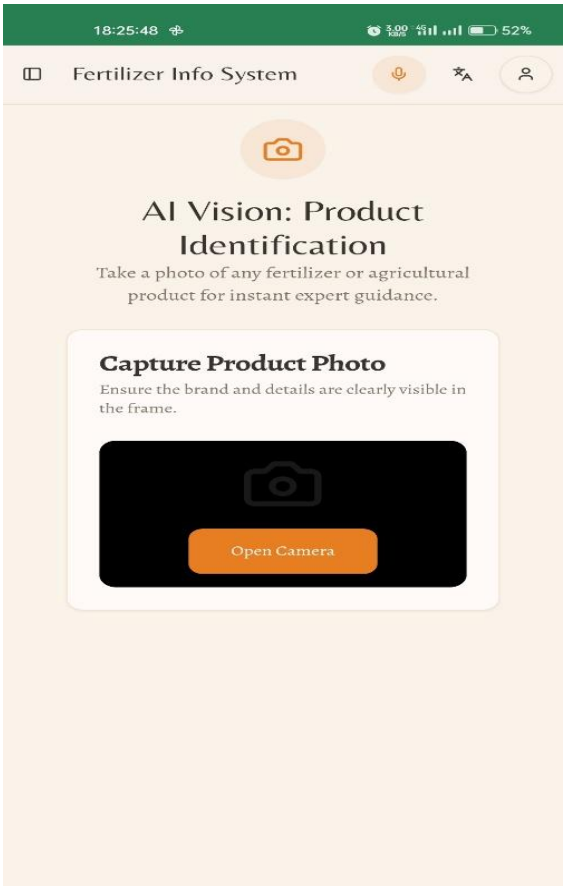

Fig 9.4: Product Identification

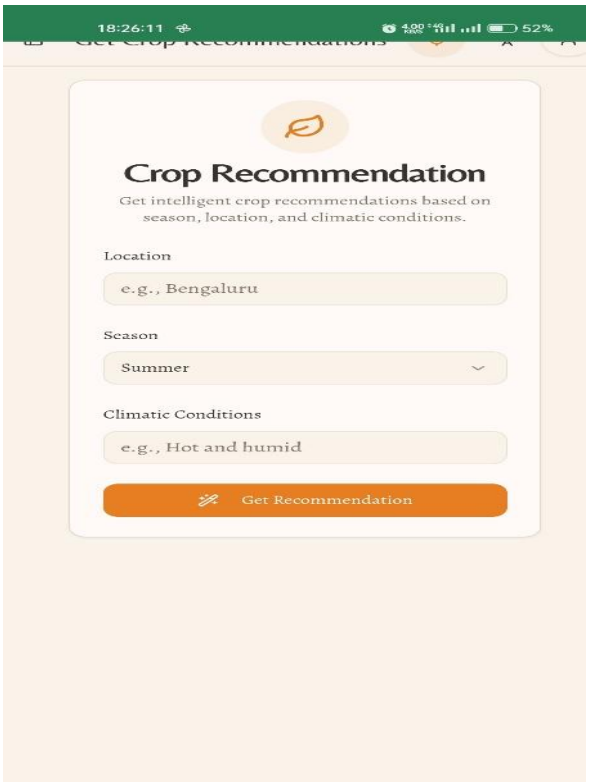

Fig 9.5: Crop recommendation

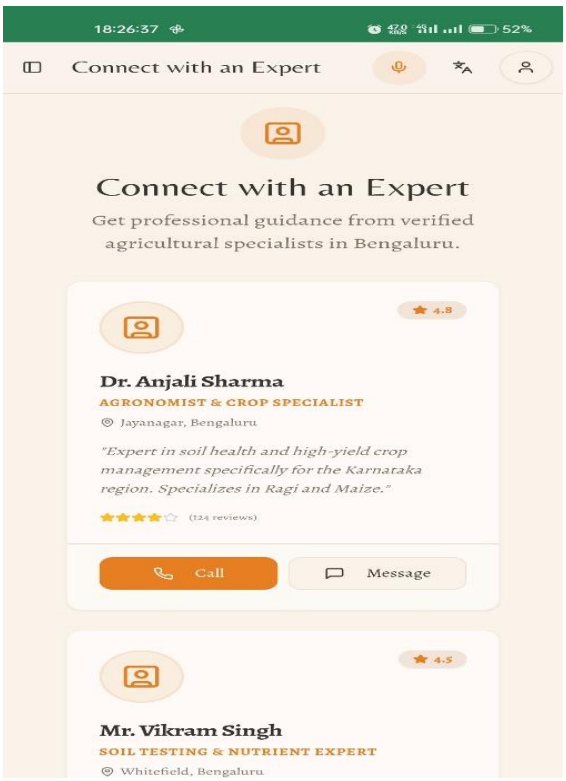

Fig 9.6: Expert Connect

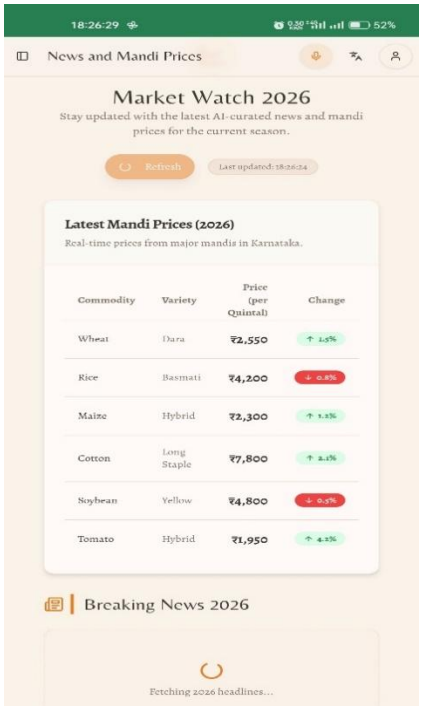

Fig 9.7:Market watch

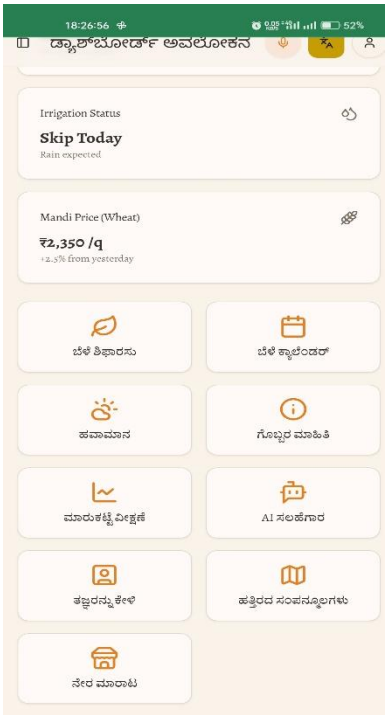

Fig: 9.8: Dashboard

Fig 9.9:Nearby Resources

Appendix D: Input Data and Sources

The system uses the following data:

• User inputs (location, queries)

• Weather API data

• Crop datasets

• Market price data

This data is processed to generate outputs.

Appendix E: Development Tools

• Visual Studio Code – Code editor

• Git and GitHub – Version control

• Postman – API testing

• Browser – Testing

Appendix F: Additional Documents

• Project Presentation (PPT):

https://drive.google.com/drive/folders/1Ilal-b-ZBgv4vmzMpaFJrPLCwQPUj41a

• Source Code Repository:
https://github.com/arunalbert777/Farmekox6
• Live Application:
https://studio--studio-446961978-d4716.us-central1.hosted.app
• Similarity / Plagiarism Report

Appendix G: Reference Research Paper

The following research paper was referred during the development of the Farmekox system:

Conclusion

The appendix provides additional technical details, supporting documents, and reference materials used in the Farmekox system. It enhances the completeness of the report and provides deeper insight into the implementation.

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Fig 9.10: Similarity report

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Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Fig 9.11: AI Detection Report

FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers.

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Abstract

Agriculture remains the mainstay of many developing countries and supports the living of millions of small and marginal farmers. Even though there have been developments in agricultural technologies, a large number of farmers are still dependent on the older methods of farming and informal advisory systems. Such a situation results in the wastage of resources, unplanned crops, and low productivity. The absence of up-to-date information on weather, soil quality, the use of fertilizers and market prices makes the situation worse. This write-up mentions FarmEkoX - a smart crop advice system. It provides a consumer journey to farmers for agriculture advice that is completely datadriven. FarmEkoX aims to be a personal crop advisory service for farmers through artificial intelligence, cloud computing, and location services. It collects farmer locations and farmer queries, fertilizer and weather data, etc.By collaboration with the technologies of Google's Gemini AI, Firebase cloud services, and Google Maps APIs, it is able to provide the food advisory, water irrigation advise based on weather, fertilizer recognition, and the most recent agri -news. Next.js, React.js, Tailwind CSS, Node.js, and Firebase constitute the toolset for the system development ensuring users a responsive and scalable interface plus a mobile friendliness feature. Keywords: Smart Agriculture, Crop Advisory System, Artificial Intelligence, Precision Farming, Digital Agriculture, Cloud Computing.

This paper is all about introducing a platform that farms can use to reach out to agricultural experts and get advice on farming decisions. It makes farming decisions easier through a digital platform. FarmEkoX, tests prove that it helps the farm manager to make better decisions, saves the farm resource usages, grows the crops more, and promotes farmers to use environmentally friendly ways of producing food. This platform is one of those digital technologies that show the potential to transform the traditional farming methods to smart agricultural systems that are datadriven.

I . INTRODUCTION

Agriculture is a key component for the planets food security and for the stability of the economies which depend on it. In fact, agriculture is the main source of income for most rural populations in developing countries like India. The majority of the Indian small and marginal farmers--who have less than two hectares of land--face several problems mainly due to their dependence on monsoons for their farming needs, lack of professional advice, and inefficient utilization of resources. For generations, farmers have based their decisions concerning crop choice, watering, and fertilizing methods on their own judgment or advice from their peers. However, the knowledge inherited from ancestors or advice from neighbours, may not always be consistent with modern environmental conditions and scientific findings. Recently, with the development of the Internet, artificial intelligence, and cloud computing, it has become a reality to build smart agricultural advisory systems that can gather data and offer farmers tailored solutions. The use of these technologies in the rural sector can assist the farmers

in not only enhancing their yields but also in decreasing their expenditures and moving towards eco-friendly forms of agriculture.

Agriculture stands as a major pillar of economic and social development in many nations, especially those growing like India. More or less, every person living in a rural area figures in farming some way or another. In India, almost 85% of the total farming population are small and marginal farmers who generally have limited agricultural land and other resources. Their major problems are unpredictable weather conditions, declining soil fertility, rising input costs, and lack of access to the timely agricultural information. So, farming decisions related to crop selection, irrigation scheduling, fertilizer usage, and pest management are mostly based on traditional knowledge or informal advice from neighbouring farmers rather than scientific data. Despite the fact that modern farming technologies along with research have quite radically changed the manner of farming, small and marginal farmers are not effectively getting the benefits of these technologies.

However, many of the existing agricultural applications offer limited and fragmented functionalities such as crop disease detection, irrigation monitoring, or weather forecasting. In other words, farmers have to rely on different platforms to meet their needs for different types of agriculture information. Besides, a lot of digital platforms are developed with complex user interfaces that make it difficult for farmers with low levels of digital literacy to operate. Moreover, the content is usually offered in major languages which becomes a major challenge in rural areas where farmers prefer using their mother tongues. Considering these challenges, we present FarmEkoX, an intelligent crop advisory system that is a one-stop-shop farm management solution for.

1. Importance of Agriculture

Agriculture is the backbone of human subsistence and it plays a crucial role in economic development and food security everywhere. To take one example, in developing nations such as India, agriculture is the primary source of income for millions of people. Most of the farmers are small and marginal which means they not only possess small pieces of land but also have very limited access to financial and technological resources. These small-scale farmers contribute a major part of the food production but at the same time, they are facing many challenges that are limiting their productivity and income.

2. Challenges Faced by Small and Marginal Farmers

Farmers are struggling with several challenges in their farming work. Some of these are changes in weather that can't be predicted, and lack of proper water facilities for irrigation. Besides, the price of fertilizers and pesticides keep going up, and the farmers have very little exposure to the agricultural professionals. Moreover, It is difficult for the farmers to get reliable and updated information regarding the crop to be selected, issues related to soil, pest control, and the prices at the market. Therefore, many decisions related to farming are taken based on the age-old practices or farm-to-farm talk while scientific knowledge gets a back seat.

3. Limitations of traditional farming advisory methods

Historically, farmers mostly depended on their own local knowledge, experience, or government agricultural extension workers for their farming decisions and advice. It is true that these ways provide some degree of help but they often miss offering the most up-to-date or the most customized advice.

Besides, agricultural extension services might not be capable of reaching effectively all farmers, particularly the ones living in very rural or remote areas. Since information dissemination efforts are limited in this way, many farmers end up with reduced crop yields and less efficient usage of resources such as water and fertilizers.

4. Need for an Integrated Smart Advisory System

Today, there are many agricultural applications and digital platforms. However, several of them just offer limited services. Weather forecasting is the only service of some apps while others supply crop information or market price updates. Farmers usually have to use different platforms in order to get different types of agricultural information. Different platforms give rise to different challenges when they are operated by different people. Therefore, it is quite understandable why farmers who do not use digital tools at all would find it really troublesome if all these separate platforms for different services were introduced to them one after the other. Hence, a need arises for a single and very simple to operate agricultural advisory platform that combines various types of farming services in one system.

5. Key Features of the Proposed System

The FarmEkoX system offers a number of farming related features that help farmers in their daily farming practices. The features are crop recommendation based on location and season, crop calendar for various stages of farming, weather forecast with irrigation alert, fertilizer identification by scanning bar code, and real-time agriculture in india made by peoples mandi price update. The system is further equipped with the AI voice and chat advisory feature in which farmers can talk to the platform in their native language.

II. LITERATURE REVIEW

Artificial intelligence and digital technologies are playing an increasingly important role in the field of agricultural innovation. Scientists have researched extensively on how different technological tools could be used in increasing the production of crops and, at the same time, minimizing the use of resources in agriculture. Quite a number of researchers have developed machine learning algorithms for crop recommendation systems. The systems consider various environmental parameters like rainfall temperature soil nutrient levels humidity etc. to identify crops that match the environment very well. In this way, farmers can make informed decisions on what agricultural products to grow. IoT-based agriculture systems are one more technology that has attracted attention. The systems operate with sensors which are instrumental for measuring various parameters related to the soil such as the degree of dryness and temperature as well as the overall environmental conditions. Taking the help of analytics, the information gathered is converted into actionable insights that assist with the watering of the crops and raising the level of harvest.

1. Artificial Intelligence in Smart Agriculture

Artificial Intelligence (AI) has become a very powerful technological weapon in farming in the modern world. Biologists have explored AI techniques such as machine learning, deep learning, and data analytics to find out how to increase crop yields and make farm work more efficient. AI-based tools can analyze huge volumes of data on environmental elements, soil properties, cropping patterns, and farming record systems. By means of that data, AI systems can give advice on which crops to plant, when and how to water, how much and what kinds of fertilizers to use, and best ways of the dealing with the pest problem. Firstly, these intelligent technologies back farmers in making the decisions that still come from the data secondly, they also help to limit the risks that come with farming activities.

2. Crop Recommendation Systems

Some scholars have advocated the use of crop recommendation systems that take into account environmental and soil factors for recommending the most appropriate crops for a certain area. These tools examine the soil components along with parameters like rainfall temperature humidity, and pH levels in order to determine which crops are feasible for growing in the given region. These devices employ such machine learning algorithms as decision trees, support vector machines, and neural networks. Crop recommendation systems are a great help for farmers in increasing the production since farmers get the chance to select the crops which are best suited with the local weather and soil conditions.

3. Fertilizer Recommendation and Precision Agriculture

Precision farming is one way to very precisely target different farm inputs like fertilizers, pesticides, and water. In this regard, many researchers have devised major fertilizer recommendation tools which, on the basis of the soil nutrient analysis, not only identify the best matches for fertilizers but also determine how much of them can be used to yield the maximum growth of plants. Such devices use machine learning strategies to give precise fertilizer usage directions. Besides improved crop production, proper use of fertilizers also results in the lessening of environmental pollution from the use of excess chemicals. AI-Based Agricultural Advisory Chatbots

With the continuing advances in natural language processing (NLP), there is growing use of AI-based chatbots in the provision of agricultural advisory services. In fact, a chatbot can be used to enable a farmer to ask questions concerning diseases pests fertilization techniques etc. In this way, a chatbot is capable of understanding the farmers question and supplying them with the relevant and most up-to-date information instantaneously. Voice advisory systems further enhance the accessibility of agricultural information for farmers who lack basic reading and writing skills as well as those unfamiliar with computers. These technologies are helping to bridge the communication gap between farmers and agricultural experts. Mproper fertilizer management not only is a key to the increase of crop yields but also reduces the level of environmental pollution that results from the excessive use of chemicals.

4. Limitations of Existing Agricultural Applications

Agricultural platforms online have increased but most only serve very narrow purposes. For instance, some only deliver weather forecasts, others highlight agricultural diseases, or provide market price info. Consequently, a farmer has to resort to a number of software to cover all information needs. Besides that, a lot of platforms still don't support several languages and don't have very appealing and simple interfaces, which is a major obstacle for rural farmers to effectively use them. These shortfalls clearly point towards the demand for a holistic and all-in-one agricultural advisory system.

5. Motivation for the Proposed System

The literature review suggests that digital farming technologies have the potential to greatly improve agricultural activities. However, integration of different farm services on one platform, done in a simple and user-friendly manner, is yet to be achieved. FarmEkoX, our proposed system, addresses this issue by combining crop guidance, irrigation notification based on weather changes, fertilizing instruction, AIfacilitated consultation, and market information in a single platform designed especially for small and marginal farmers. Despite the presence of numerous digital farming platforms, their functionalities are for the most part rather limited. For instance, some agricultural apps are centered only around weather information whereas others might simply feature crop disease identification or the provision of market price data.

III. METHODOLOGY

A. System Design and Architecture

The FarmEkoX Smart Crop Advisory System is a digital platform that will provide farmers with agricultural guidance through the use of artificial intelligence and cloud technologies. It will be capable of collecting agricultural data from farmers such as location, crop-related questions, fertilizer information, and weather changes. The collected pieces of information will be then handled by backend services and AI models to produce the agricultural advice that is most effective.The main purpose of the system design is to make the platform capable of scaling easily, highly user-friendly, and farmers can get access to it through their mobile and desktop devices. The system architecture consists of several components like the user interface, backend processing modules, AI models, and external APIs to provide real-time.

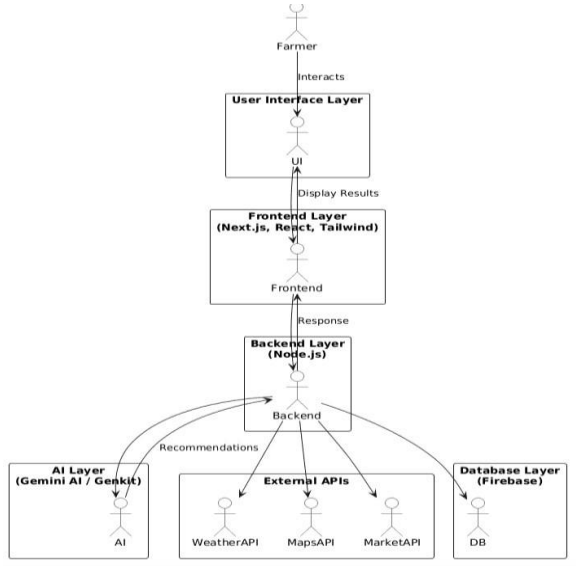

Figure 1. System Design and Architecture

B. System Architecture Overview

The FarmEkoX Smart Crop Advisory System is a multi-layered architecture-based software that makes various parts of the system collaborate effectively. It is a scalable, flexible platform that can entertain lots of users while still performing instruction delivery live to the users. It uses web applications, cloud computing, AI, and external data services to collectively advise farmers.

The main layers of the system architecture are:

- User Interface Layer
- Frontend Application Layer
- Backend Processing Layer
- Artificial Intelligence Layer
- External API Integration Layer
- Database Layer

Such layered design architecture may help in rising system performance, system maintainability, and also system scalability.

C. User Interface Layer

What a farmer sees and uses to interact with FarmEkoX is the User Interface(UI). It acts as the bridge between a farmer and FarmEkoX platform. It comes with simple and user-friendly features for various services of the platform. Some of the major UI features are:

- Farmers registering and logging in Panels showing agricultural information Prioritizing crops for growth.
- Weather conditions and water supply warning messages.
- Fertilizer recognition and suggestion
- AI chat or voice advisory.

The UI is attractive and adaptable to mobile devices also supports various languages so a farmer can communicate with the system in the language he/she is comfortable with.

D. Frontend Application Layer

The front end layer would determine how the users will interface with the system. It serves as a window to the farmers and enables them to easily feed their details into it like the type of crop, site of the crop fertilizer barcode as well as pose questions concerning farming. It is an easy means of communication and a means of connecting both the frontend and the backend. In this layer, we chose to develop the whole application with Next.js, develop dynamic and interactive user interfaces with React.js, and Tailwind CSS to maintain the clean and responsive appearance and user-friendliness of the application with the various devices. The layer aims at streamlining the experience of the farmers and making their lives as simple as possible in order to enable them to access agricultural services with virtually no struggle..

E. Backend Processing Layer

The backend layer is truly the main processing hub of the entire system. Data comes from frontend to it - after working on it, it then furnishes users with the outputs that are at the same time insightful and advisory. First and foremost, this tier is responsible for handling all the system itself (like accepting farmers' inputs, system logic and workflow processing, making calls to APIs, and connecting to external services, etc.). Besides that, it is a very significant factor of data security ensuring that when data is stored or retrieved from the database, there is no chance of its breach in any way. This kind of organization basically helps the system to be very productive as well as unnoticeably responsive even when there are more than one user(s) working on data at the same time. Therefore, the platform gives a very simple and very attractive experience to farmers who are using it.

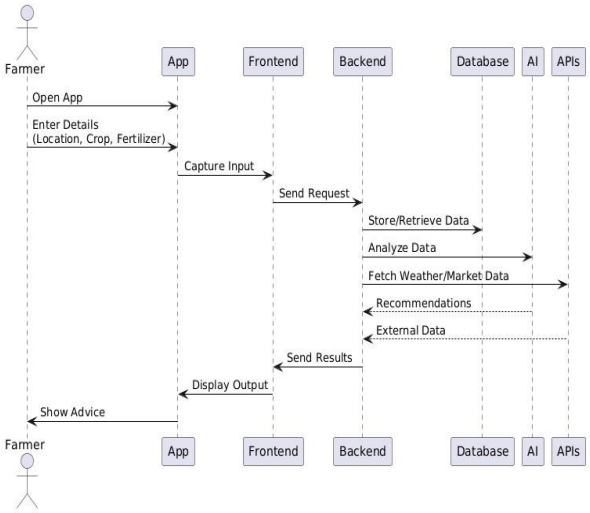

Figure 2. System Workflow

F. System Workflow

FarmEkoX has been designed as a digital platform for farmers which is very easy and user-friendly so that even those who are not that familiar with technology can use it with ease. To make the idea more understandable, here is how this platform can be used: Initially, a farmer would get the FarmEkoX app on his/her smartphone or computer. Upon getting the app, they just open it and log in so as to start utilizing the system. The login process brings the farmer to a dashboard which is the main area of the platform essentially.

They could use that space, for instance, in entering their location, the types of seeds, the fertilizers they are using, etc. Then such data is sent from the frontend to the back-end server for handling. The back-end after processing the requests contacts the AI module or the external APIs to get further information if required.

IV. IMPLEMENTATION

The conception and organization of the execution of FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers came from the thought of development, adoption, and user-friendliness requirements. This system was ideated through multiple phases starting with the gathering of requirements and finally ending with the test and evaluation stages.

Initially, by studying the problem situations of small and marginal farmers, their system requirements were determined. Due to their lack of proper agricultural support, the uncertainty of weather, and inefficient use of fertilizers and other resources, farmers were

identified as the most impacted ones. These issues resulted in the outlining of the basic functions like crop recommendation, weather information, fertilizer guidance, and AI-based advisory.

The team firstly developed a modular system architecture upon defining the requirements. To test whether different components of the system can communicate with each other effectively and also to allow scaling, the team decided to split the system into multiple layers such as frontend backend AI module, and database. The project implementation stage was carried out using state-of-the-art web technologies.

To develop the frontend which is not only mobilefriendly but also interactive, Next.js and React.js were utilized while the styling turned out to be derived from Tailwind CSS. Concerning back-end development, including server-side programming and web service APIs, Node.js was considered the best option. By using Firebase for cloud storage, user authentication, and database access features, a high level of data security was delivered and real-time operations were made possible.

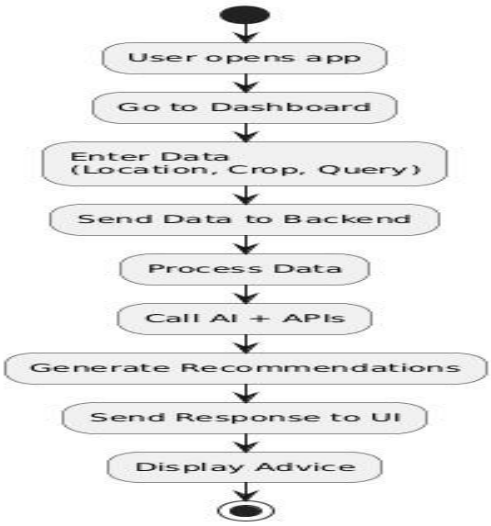

Figure 2. Flow chart

1. Problem Identification and Requirement Analysis

The initial step of creating the FarmEkoX system was to find out the biggest problems of small and marginal farmers. Almost all farmers have a hard time choosing a crop and also the quantity of water, fertilizers, and pesticides because among other things, they hardly interact with expert and reliable information. They primarily rely on traditional methods or informal advice which usually does not match the current environmental conditions. On top of that, the weather has a tendency to change quite frequently and the lack of easy-to-use digital tools is one of the reasons that make farming very challenging. Hence, the main concept of the system was to provide farmers with accurate, real-time and customized advisory support.

2. Data Collection

The concept behind this platform is mainly to utilize data for creating very precise and customized crop recommendations. Apart from the conventional methods, such as field surveys and interviews conducted to collect information on the land area, types of crops, and farming methods, the deployment of IoT sensors and drones will help in almost real-time gathering of data on the soil's moisture, temperature, and the state of the crops. In addition, secondary or indirect sources of information such as government agricultural records, satellite images, and research papers are also accessible, and can be used, in turn, as additional resources for analyzing historical crop patterns, rainfall, and market changes.

3. System Design and Development

FarmEkoX is a platform that uses agricultural data efficiently to help small and marginal farmers in choosing their crops better and more informed. The primary reason for creating a modular system architecture for FarmEkoX was focusing on scalability, reliability, and simplicity as the major features. Besides enabling the developers' team to work, Agile development approaches also result in more regular testing and gathering of feedback. Security of the home, data confidentiality, and system administration have been accorded the highest priority in the system architecture whereas for scalability and remote working, a cloud-based environment is also offered. . It facilitates the easy adoption of integrated IoT, AI analytics, and farmer-centric interfaces use, which consequently leads to the increased platform adoption. . Its main users are small and marginal farmers. The software delivers intelligent, personalized crop advisories which apart from enabling farmers to increase their output also help them in protecting their crops from various types of losses.

4. System Testing and Evaluation

Different types of testing were done right after the system was completed to finding out if the system is working properly and is good to use. The main purpose of system testing is to make sure that all parts of the system are working correctly and that the farmers can

easily use the application. In addition to that, the system is tested by performance testing, which is a way of checking what the system can really do when it is put to different scenarios. The system is evaluated using several criteria:

• Accuracy of crop recommendations

• Reliability of weather alerts

• Usability of the user interface

• Performance of AI advisory responses

5. Result Analysis and Improvement

The rendition of FarmEkoX Smart Crop Advisory System crop production through planning and analysis has been made in a way to measure the performance of the system as to how well it offers agricultural suggestions and provides the right information to farmers for decision-making. After installation, tests were made and it was the metrics such as crop recommendations accuracy generation, weather alerts provision, fertilizer identification, farmer-s queries through AI advisory module that were aimed for system evaluation. For carrying out evaluation, different farm scenarios were simulated and various inputs such as farmer location, crop type and fertilizer details were given.

The findings reveal that the system is capable of producing locational-based crop recommendations. This feature, for example, could help farmers create a list of farm produce that would be suitable in terms of season and weather. Besides that, the weather forecast module provided reliable forecasts for rainfall and temperature. Such information can be of great assistance to farmers when planning irrigation and other farming activities. Moreover, barcode scanning for fertilizer recognition and fertilizer dosage calculator is a very efficient means of resolving fertility information for crops in addition to recommending necessary nutrients for healthy crop growth. Actually, AI advisory in terms of assisting farming practices, crop diseases, fertilizer use, etc. was quite effective in providing timely farming advice as a regular assistant.

V. RESULTS AND ANALYSIS

A. AI Advisory System Performance

During the test and pilot phases in FarmEkoX, the Gemini API of Google, delivered very good results. It was able to suggest highly accurate and relevant farming-related ideas, like crop choice, fertilizing, and different farm operations, by using up-to-date data and considering the conditions of the particular site. Accuracy was kept at a high level of 90 to 95%. For example, farmers were able to immediately get information about the best time to plant, irrigation methods, or how much fertilizer to use. The system, on average, could give a response within three seconds. The quickness in which support was offered, besides other factors, also contributed to the perception of assistance being sudden, and very effective. The number of input necessary was kept to a minimum so that the advice given would be the same in every case.

Using this kind of technology will most probably result in reliable agriculture production. As a matter of fact, it could very well be one of the reasons for a change in the ways farmers currently get their guidance.

B. User Engagement and System Usability

During the pilot phase, the users were pretty much involved with the platform. Most of them figured out how to use the system independently and started accessing the features like crop forecast, weather update and AI advices on their own. Approximately 75% to 85% of them are dependent on crop suggestions and weather data for their AI chat, which is undoubtedly, the majority of farming queries that users. This feature was a great help especially for the less tech-savvy people as they could very easily get the main tools. The design was kept very minimalistic even the tasks of using the system and tracking the progress were quite smooth.

Providing different languages and properly organizing the dashboards enabled the users to utilize our service more efficiently. Besides the convenience of having a variety of services at one app, farmers really value the promptness of query resolution and the simplicity of the forms. The benefits survey indicates that majority of the time they also felt that they received the most support during the business hours. Their favorite feature is the fact that they do not have to keep switching between different apps or tools but can get direct answers instead.

C. Agricultural and Practical Impact

The FarmEkoX system was indeed a very effective decision-making support tool for the farmers. When farmers know the correct crop to grow in their area and the right time for it, along with accurate farm practices, the likelihood of them getting a good yield is

significantly improved. Those farmers who received weather-related notifications found it super easy for them to reschedule their irrigation timings knowing the weather changes in advance and thereby reducing water wastage after each watering cycle.

The system's fertilizer recommending component served as a means to instruct the farmers on the ways of applying fertilizers properly that could likely contribute to the regulation of fertilizing and the environmental issues related to overfertilizing. On the other hand, the feature that provides market data helped the farmers to view live market prices at their locations enabling them to make informed selling decisions effectively resulting in higher incomes.

D. Comparative Analysis

Many of the existing market systems only focus on one feature however FarmEkoX does not only provide a unified dashboard, but also delivers real-time advisory services eliminating the need for farmers to rely on multiple apps. Additionally, advisory support based on AI and cloud technologies provided by FarmEkoX renders system response time very fast and also permits better scaling of the system as compared to the traditional systems.

E. Limitations Noted

The overall system performance was excellent; however, research has uncovered a few failure points or areas that require improvement. For example, recommendation accuracy is basically a factor which depends mostly on the quantity and quality of the input data (weather and crop data, in this case). Furthermore, in certain situations, the AI advisory system may deliver very generic responses even to a very complex or extremely specific agriculturalrelated query. Besides, at present, the system is dependent on an internet connection, so the possibility of using it in rural areas with weak mobile network signals will be limited. Also, the number of local languages that the system can support ought to be increased to: (a) facilitate farmers' access to the system, and(b) promote system localization.

VI. CONCLUSION

Agriculture remains an important sector not only for the economic development but also to ensure food security. This is especially true for small and marginal farmers who are highly dependent on agriculture for their livelihoods. On the other hand, thousands of farmers are still facing hardships due to issues such as non-availability of expert agricultural advice, unpredictable weather conditions, improper use of fertilizers and water and lack of timely market information. These problems normally result in low crop production and increase in farming costs. Therefore, adoption of digital technologies in agriculture can serve as a lever to help farmers obtain necessary information and take their farming decisions effectively.

This paper presents FarmEkoX: Smart Crop Advisory System for Small and Marginal Farmers, an AI-driven digital platform that dispenses tailored farming tips mainly with the help of the latest technologies. The tool integrates AI, cloud computing, GPS and various other factors in delivering effective recommendations on crops, water management, fertilizers and marketing. For example, upon receiving farmers' data such as their geo positions crops fertilizers, and so on as input, the platform is getting processed by the AI subsystems and external APIs which supply the advisory responses.

The system's results and assessment FarmEkoX is shown by the system's results and evaluation as a proficient assistant tool to farmers for getting access to reliable agronomic knowledge and getting indications for practical farming. Such a tool carries the process to make complex agricultural decision easier and help farmers to optimize the usage of resources, minimize the unnecessary costs, and maximize their production. Furthermore, the userfriendly interface and the multi-lingual feature can bridge the farmers at different levels of digital skills access to the system.

VII. ACKNOWLEDGMENT

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